



Introduction to DAS for Seismology: From Data Acquisition to Analysis

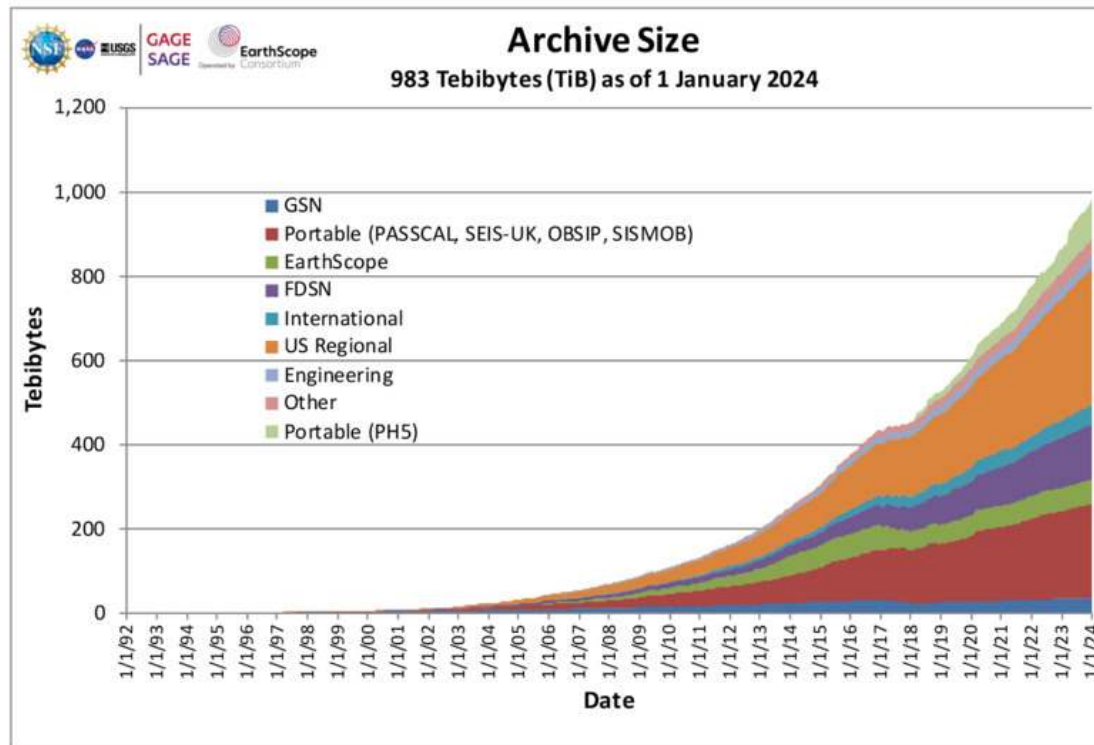
Organizers: Ettore Biondi, Jiaxuan Li, Chun Zhang, and Weiqiang Zhu
(alphabetical order)

Deep Learning for Distributed Acoustic Sensing: PhaseNet-DAS and DASNet Applications

Big Data Challenge in Seismology

Seismic Network Data

(Large-N and Large-T)

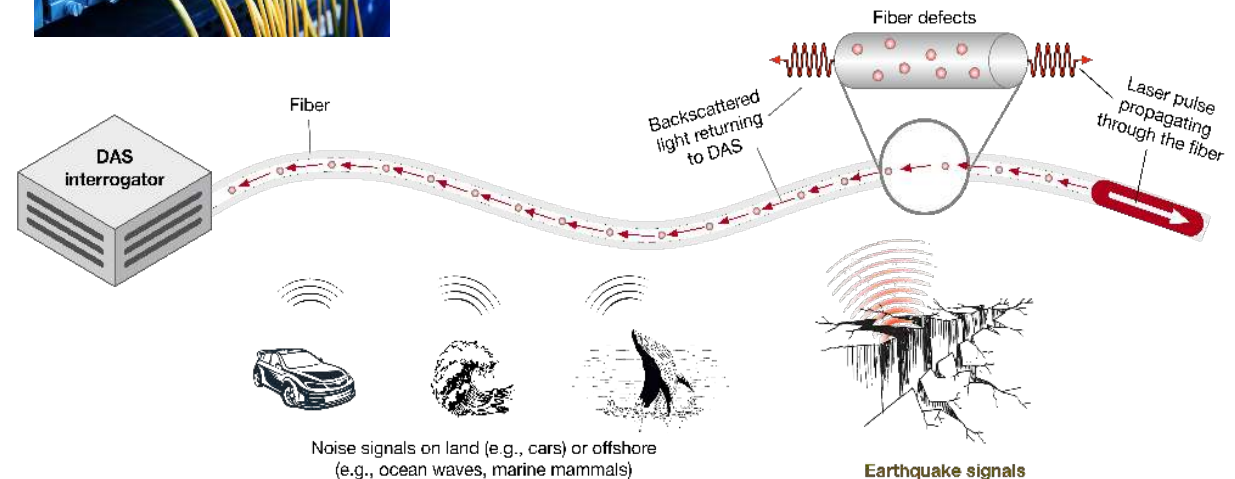


<https://ds.iris.edu/data/distribution/>

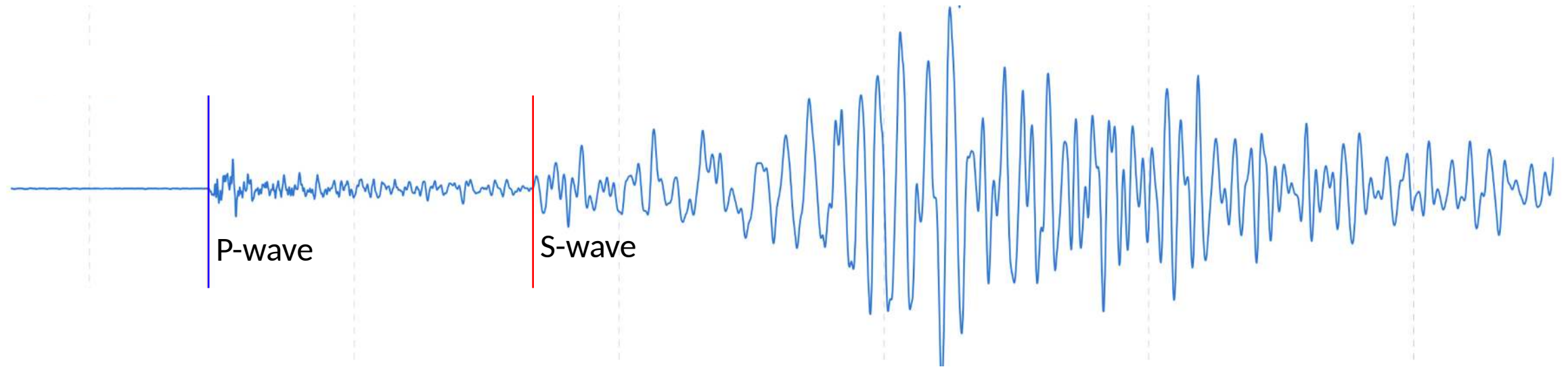
New Sensing Technology

(Terabytes of data per day)

Distributed Acoustic Sensing

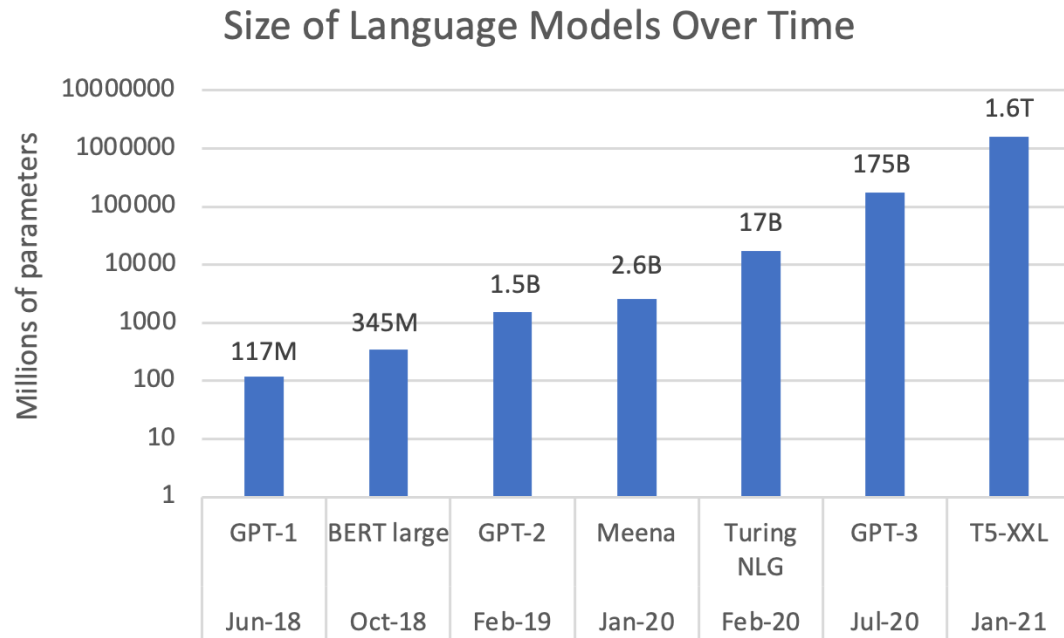


Rapid Development of Deep Learning



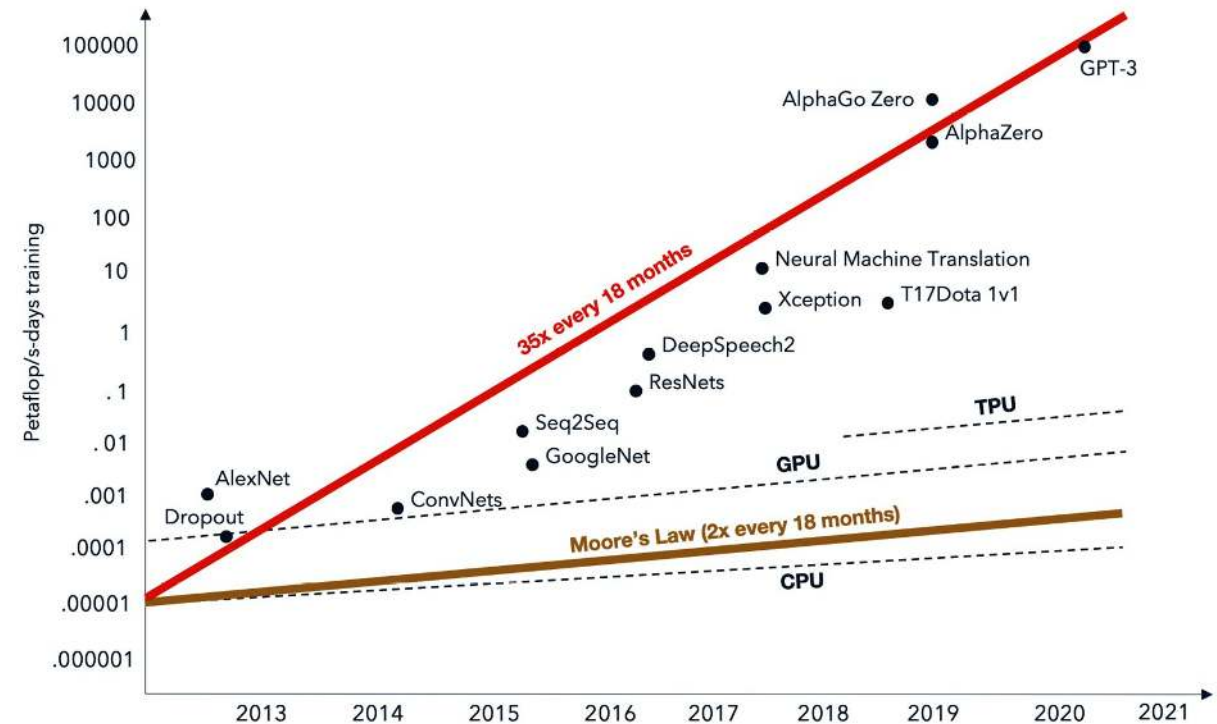
Rapid Development of Deep Learning

Performance Increase with Neural Network Size and Dataset Size



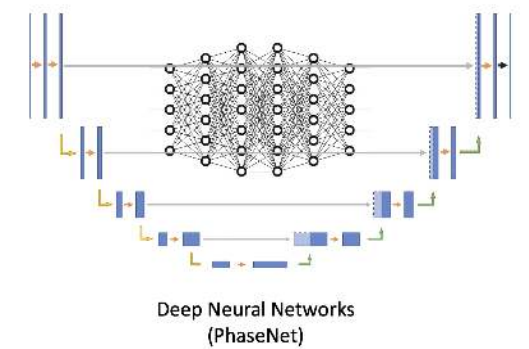
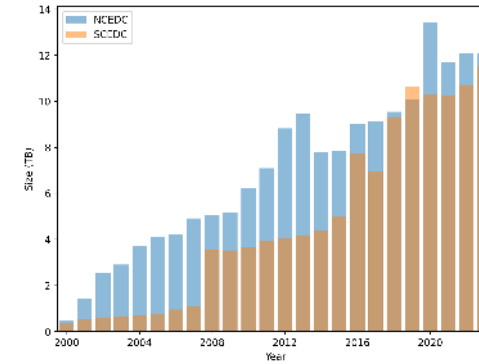
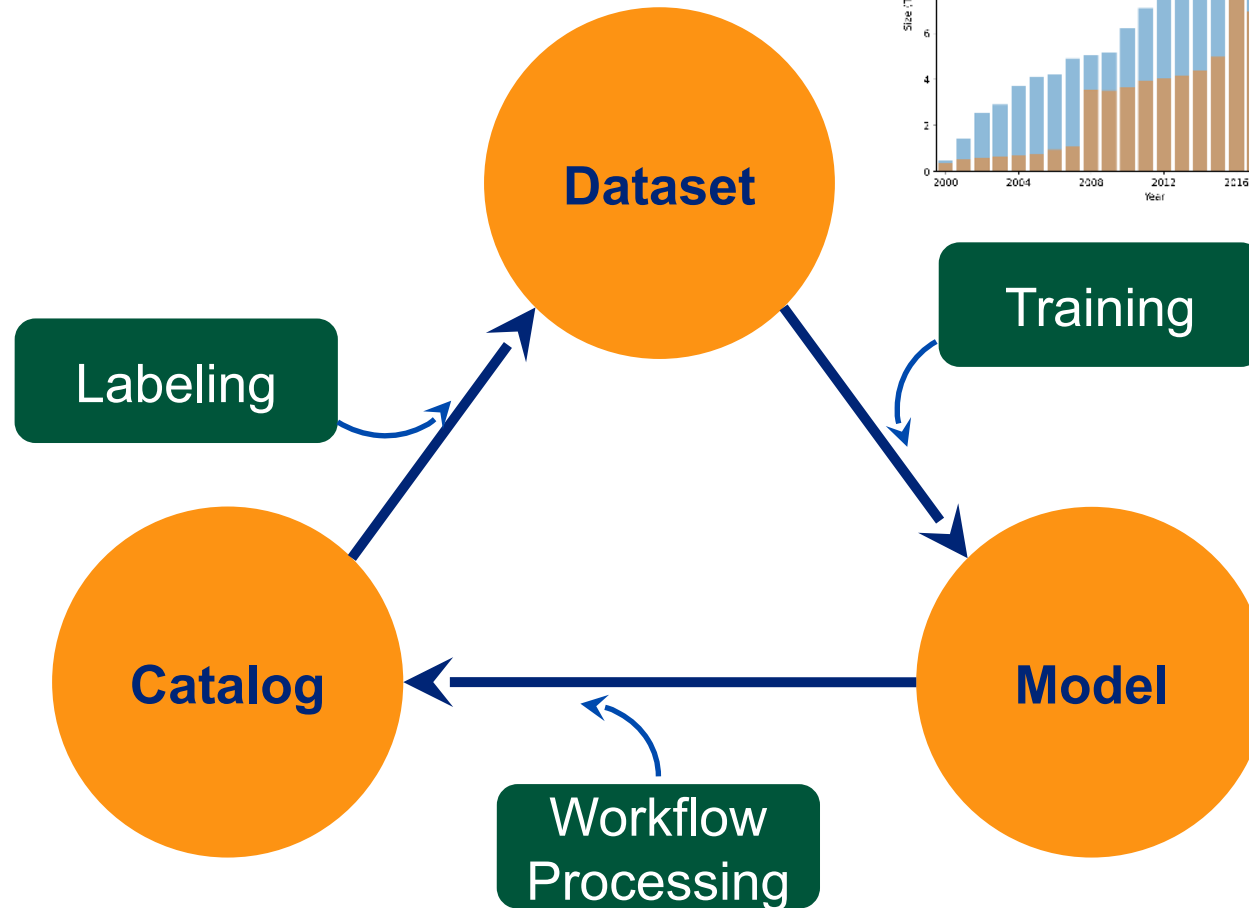
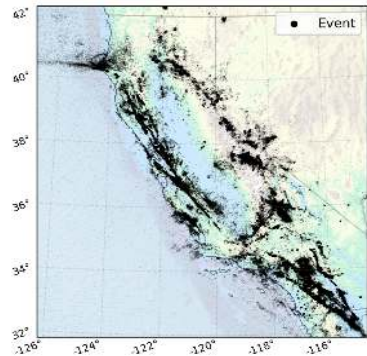
(Douwe et al., 2021)

Exponential Growth in Training Compute



(Engineering MLOps, Emmanuel Raj)

Deep Learning + Seismology

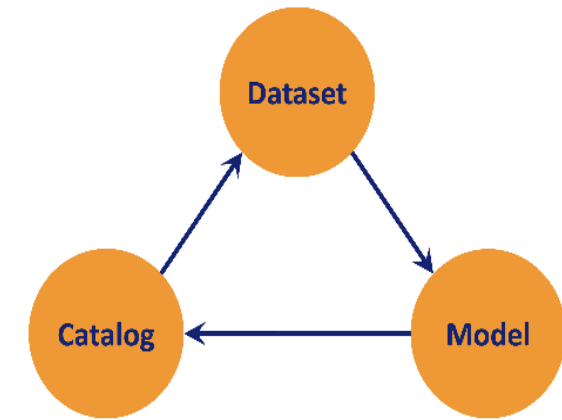
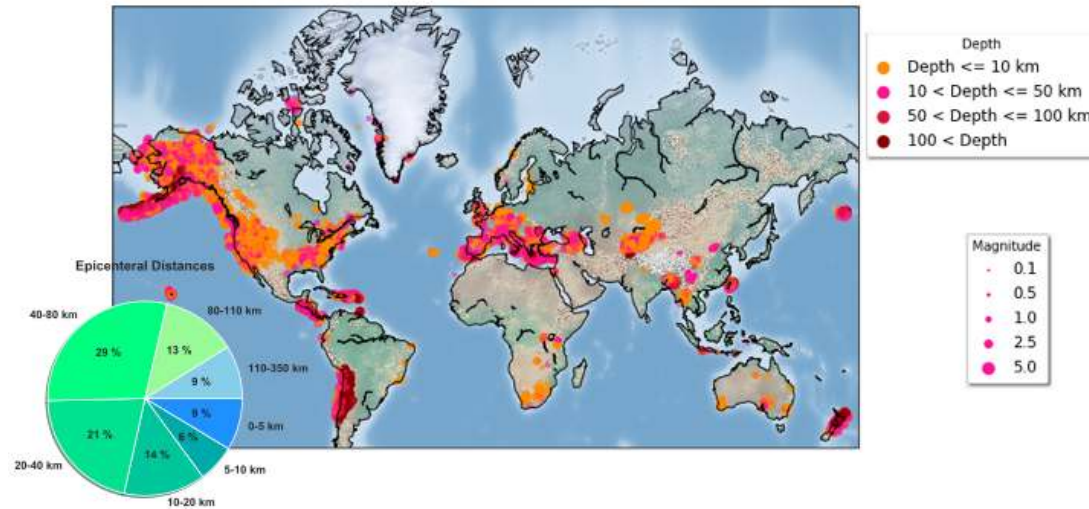


Datasets:

- **STEAD**

(Mousavi et al., 2019)

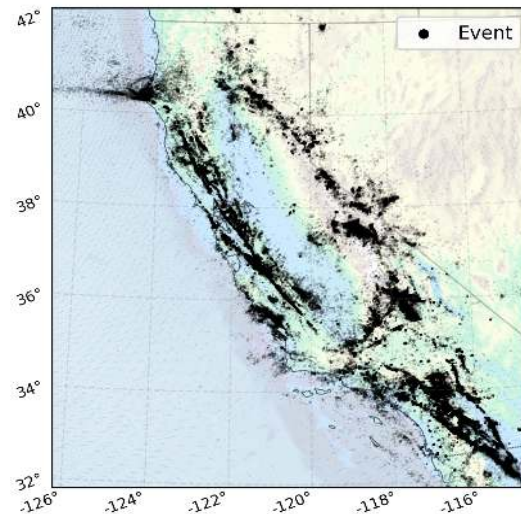
1.2 M Labeled Waveform. **450 k** Earthquakes. **19,000** Hours of Data.



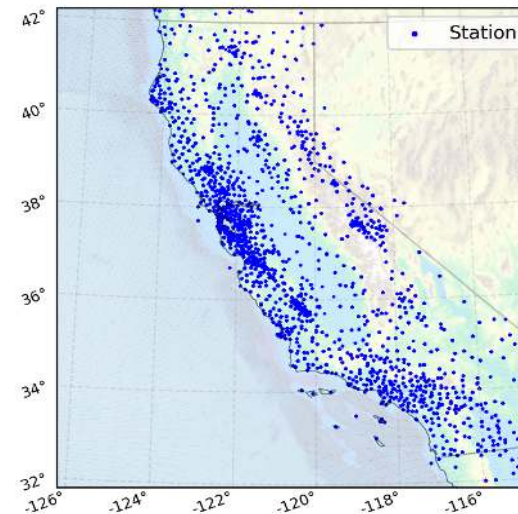
- **CEED**

(Zhu et al., 2025)

Events (~650K)

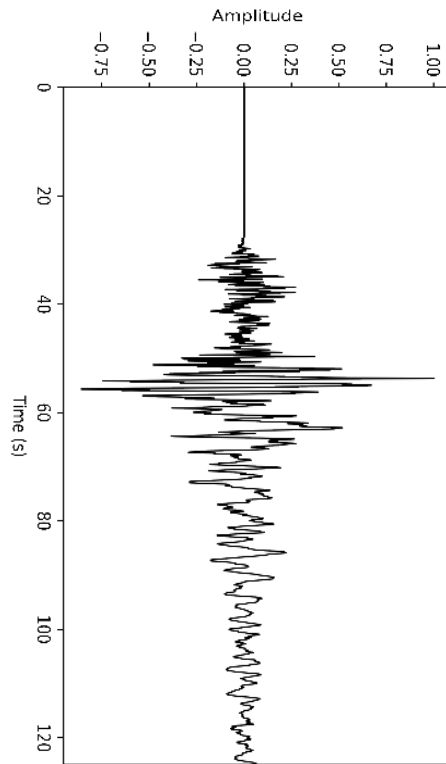


Station (~1200)

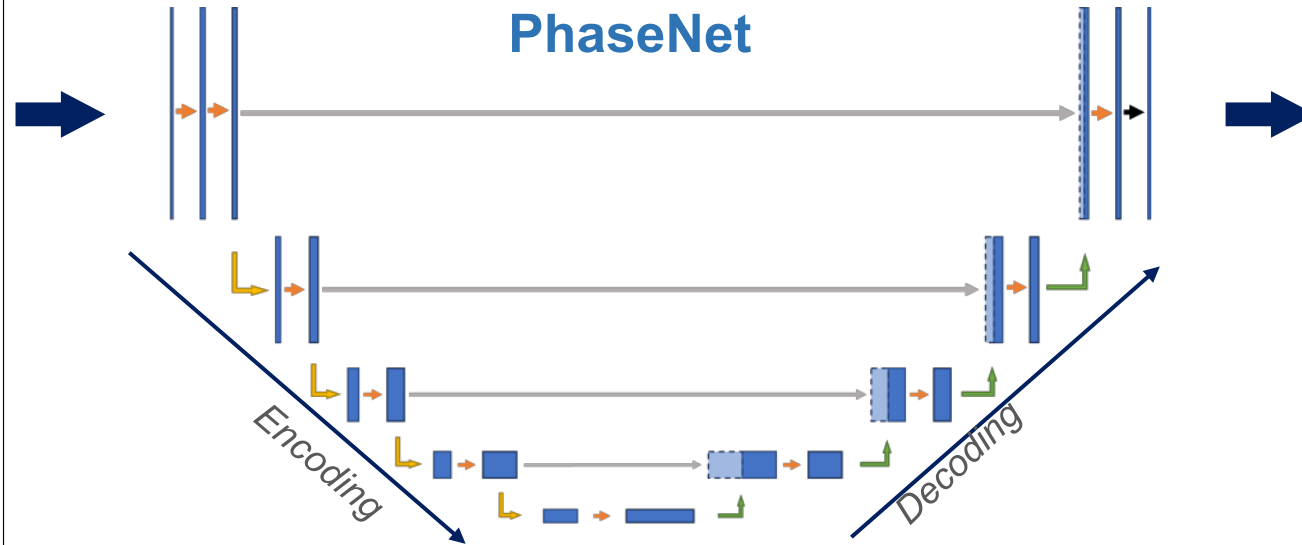
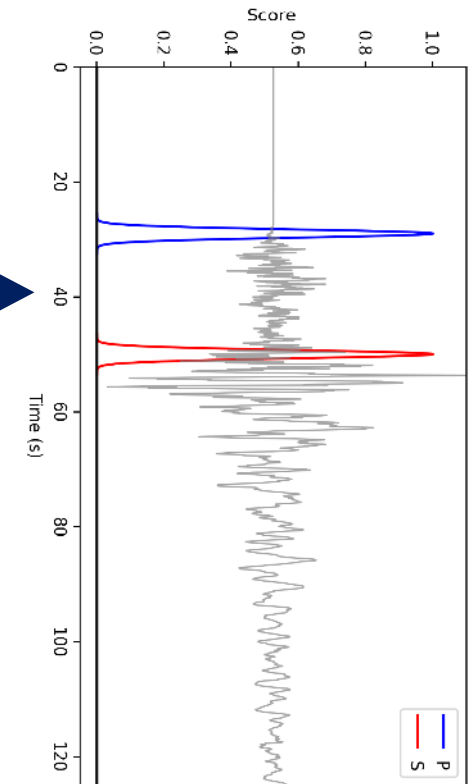


PhaseNet: Phase Picking (supervised learning)

Input Data



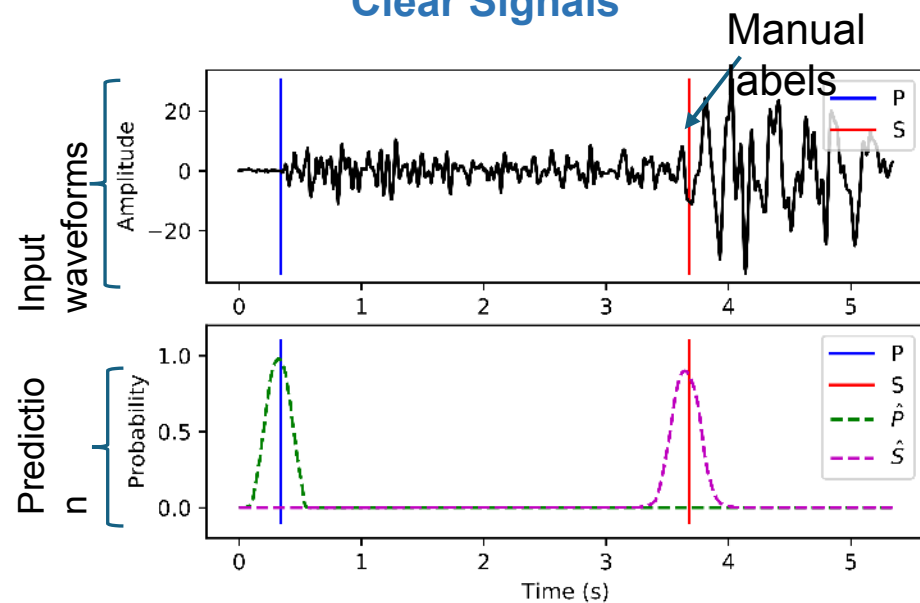
Target Function



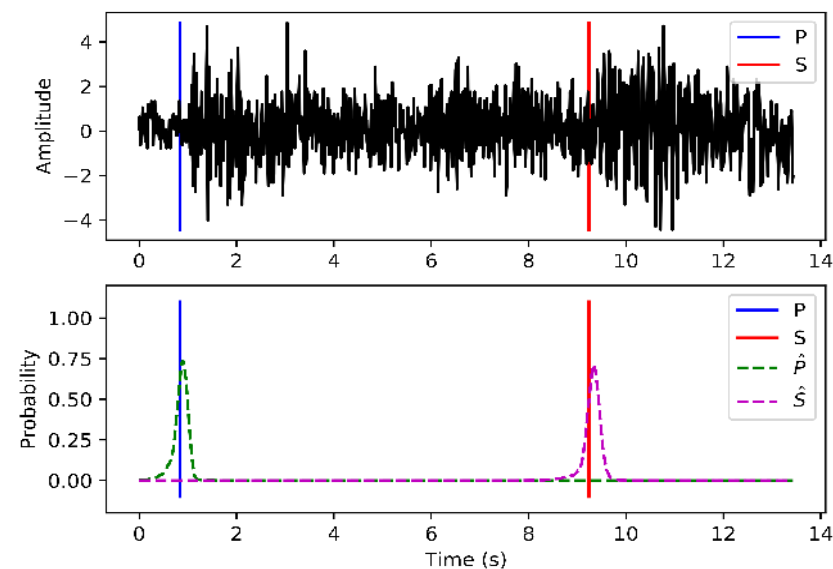
(Zhu and Beroza, 2018)

Examples

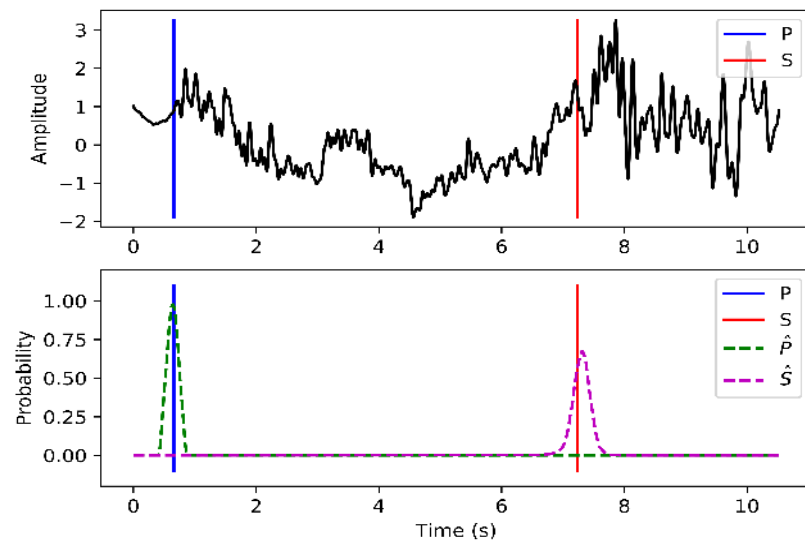
Clear Signals



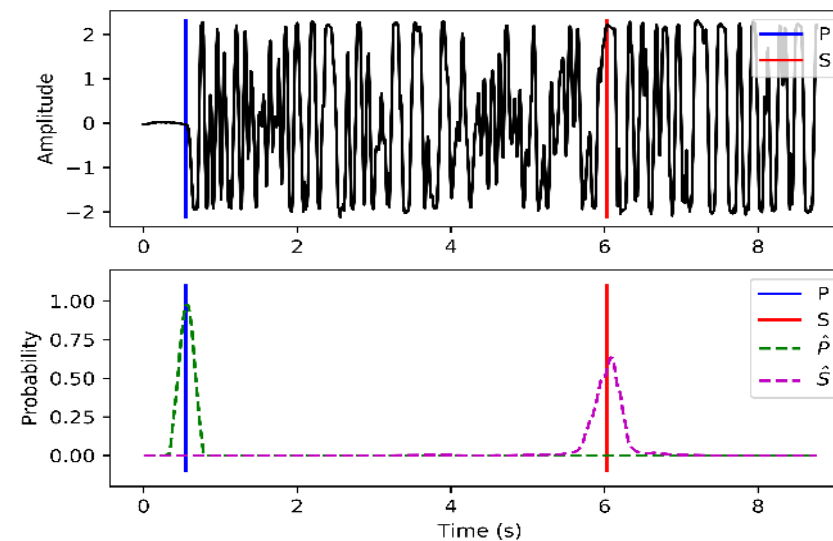
Noisy Signals



Noisy Signals

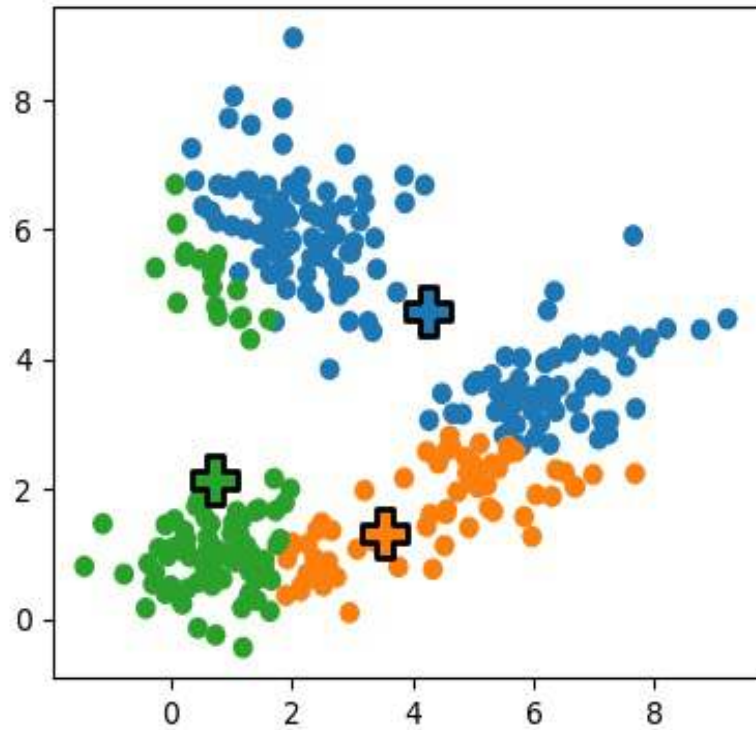


Clipped Waveforms

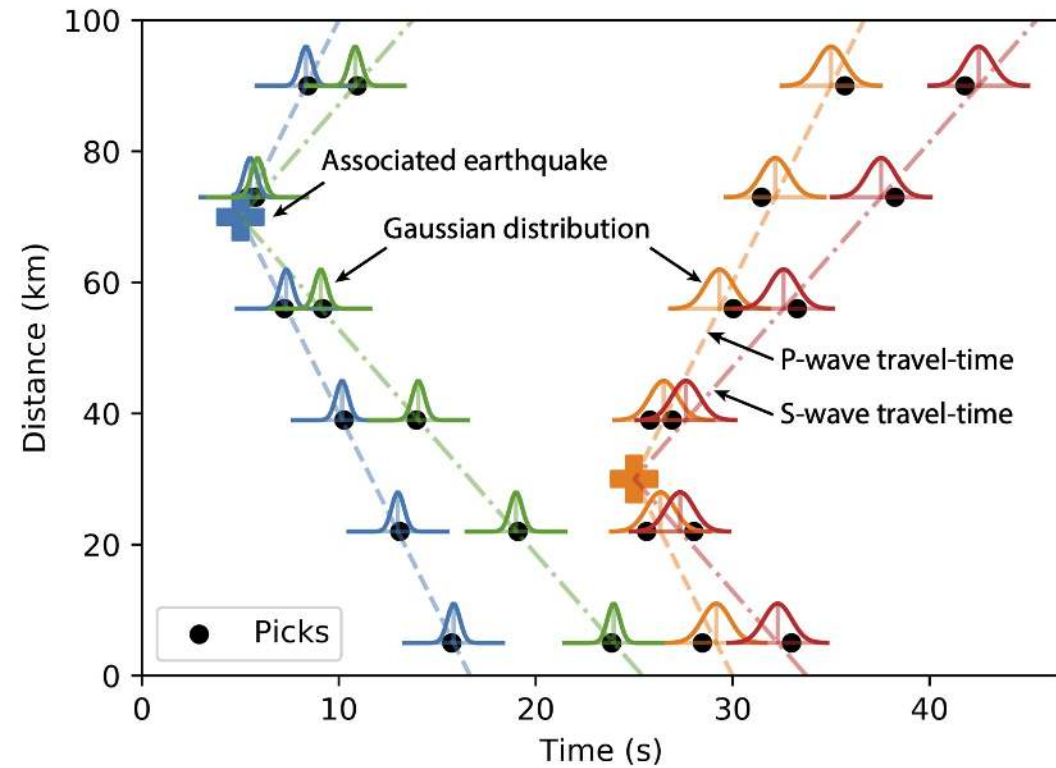


GaMMA: Phase Association (unsupervised learning)

K-Means
Clustering
Iter = 1

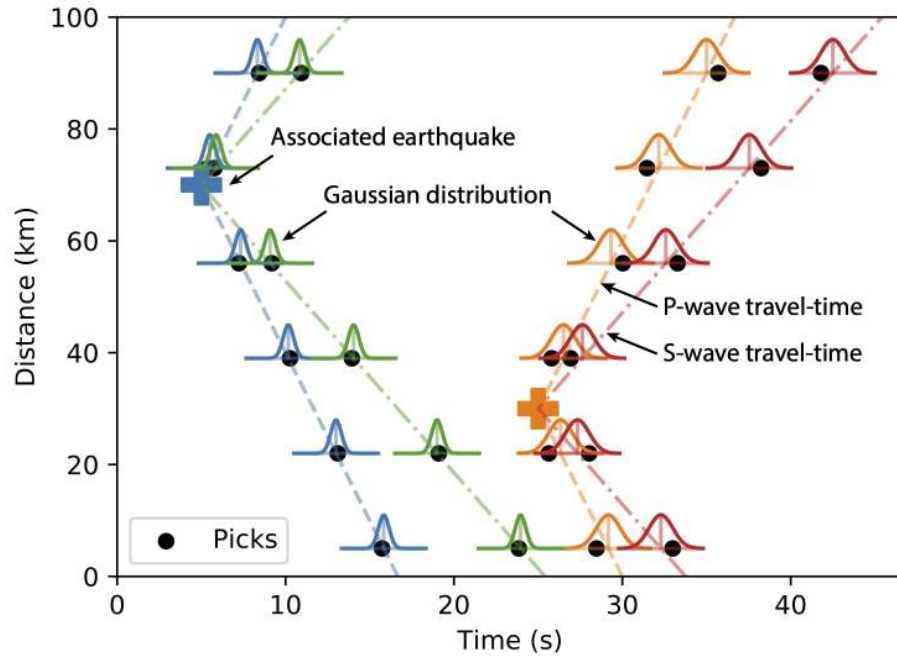


Phase
Association



GaMMA: Gaussian Mixture Model Association

- Modeling Gaussian distributions based on theoretical travel-time and amplitude
- Solving clustering using the Expectation-Maximization algorithm



(Zhu et al., 2022)

- E-step (phase space): Assign picks to each earthquake

$$\gamma_{ik} = \frac{\phi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}{\sum_{k=1}^K \phi_k \mathcal{N}(\mathbf{x}_i | \mu_k, \Sigma_k)}$$

- M-step (earthquake space): Update earthquake parameters

$$\phi_k = \sum_{i=1}^N \frac{\gamma_{ik}}{N}$$

$$\text{minimize}_{(x_k, y_k, z_k, t_k)} l(x_k, y_k, z_k, t_k) = \sum_{i=1}^N \frac{\gamma_{ik}}{2} \mathcal{L}(t_i, \hat{t}_{ik}(x_k, y_k, z_k, t_k))$$

$$m_k = \frac{\sum_{i=1}^N \gamma_{ik} \mathcal{F}'_a(a_i, d_{ik})}{\sum_{i=1}^N \gamma_{ik}}$$

$$\mu_k = \begin{bmatrix} \hat{t}_{ik} \\ \hat{a}_{ik} \end{bmatrix} = \begin{bmatrix} \mathcal{F}_t(x_k, y_k, z_k, t_k) \\ \mathcal{F}_a(m_k, d_{ik}) \end{bmatrix}$$

$$\Sigma_k = \frac{\sum_{i=1}^N \gamma_{ik} (\mathbf{x}_i - \mu_k)(\mathbf{x}_i - \mu_k)^T}{\sum_{i=1}^N \gamma_{ik}}$$

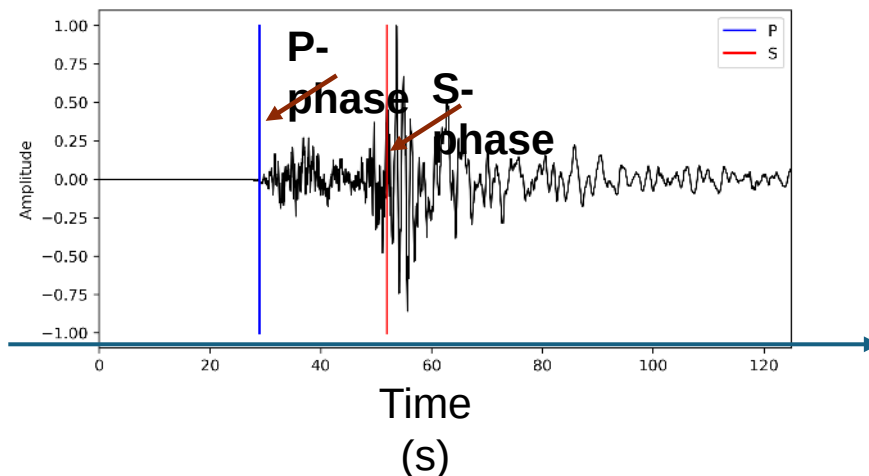
Estimate origin time
and location
Estimate magnitude
Update theoretical values
Update covariance matrix

Challenges in Applying Deep Learning to DAS

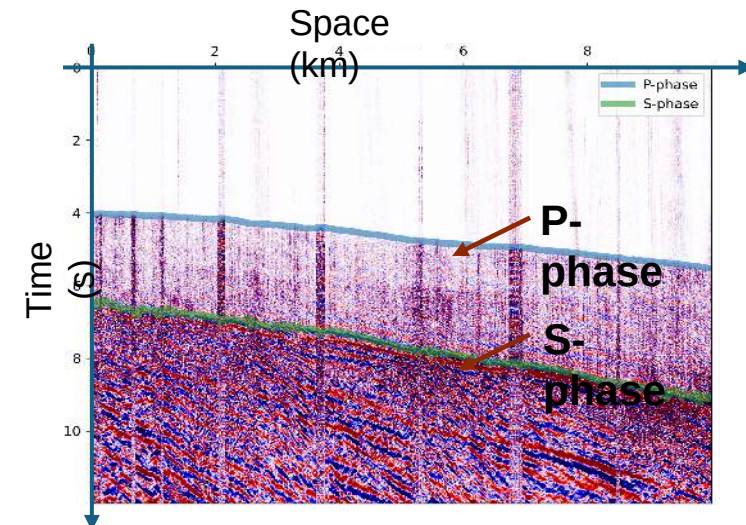
- Lack of manual labels
- Different data format from conventional seismic data

Can we transfer the knowledge from conventional seismic data to DAS data?

1D Time Series

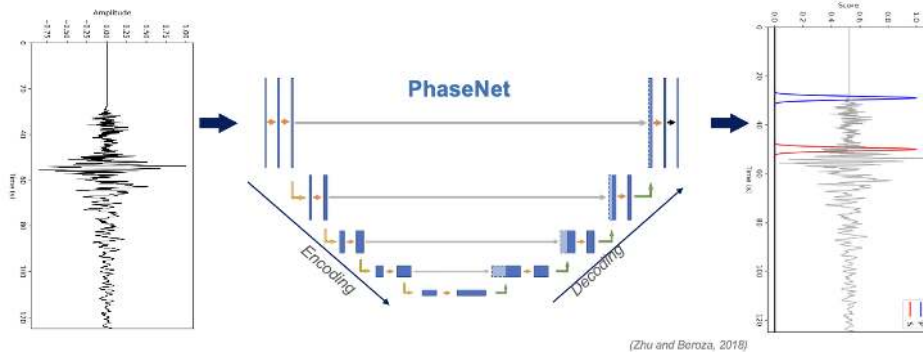


2D Spatiotemporal Data

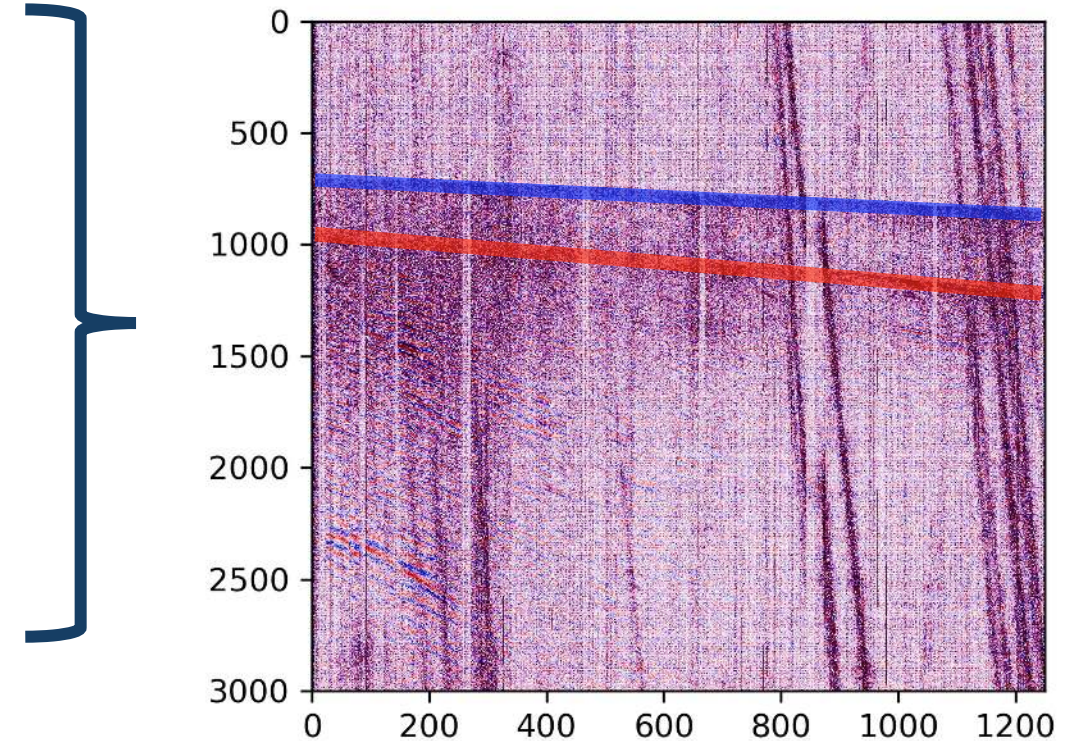
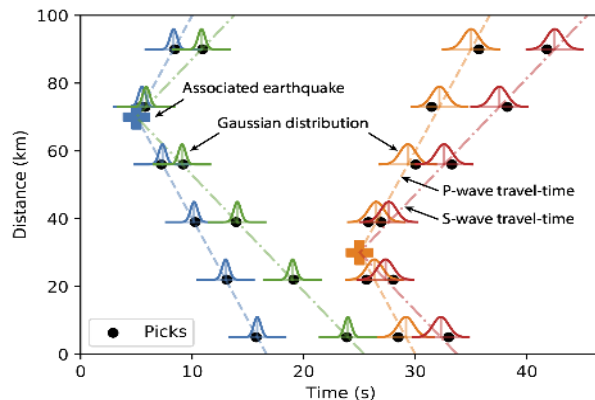


Semi-supervised Learning for DAS

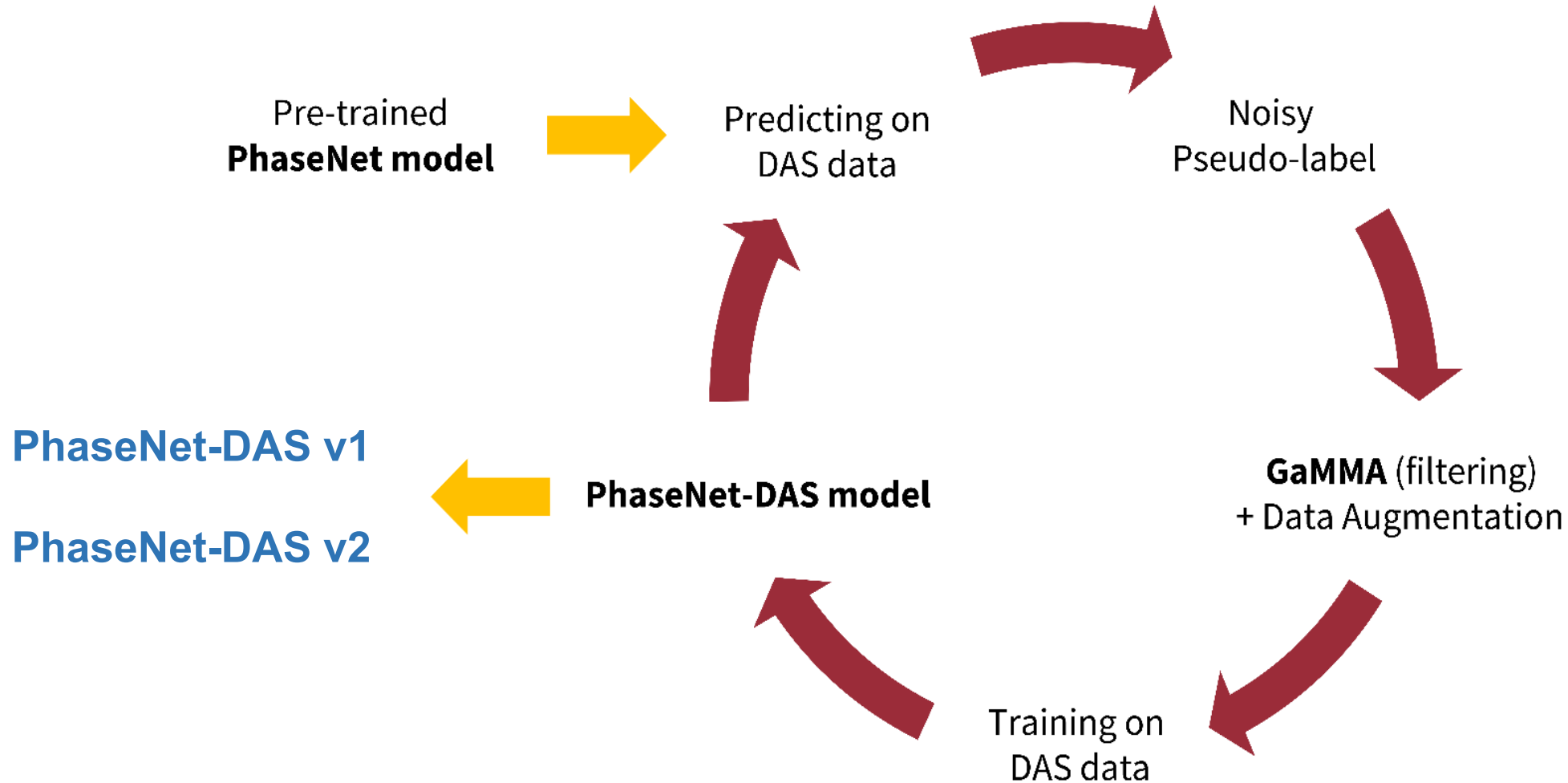
PhaseNet → Picking P&S phases in each channel



GaMMA → Selecting coherent phases across channels

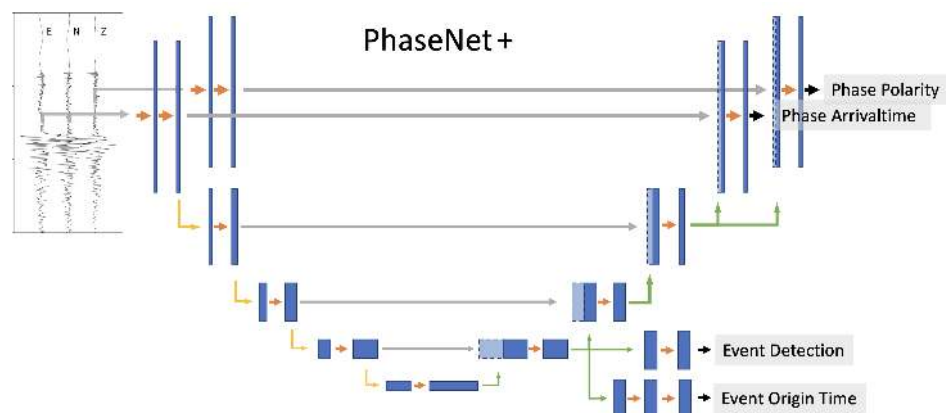


Semi-supervised learning for DAS phase picking



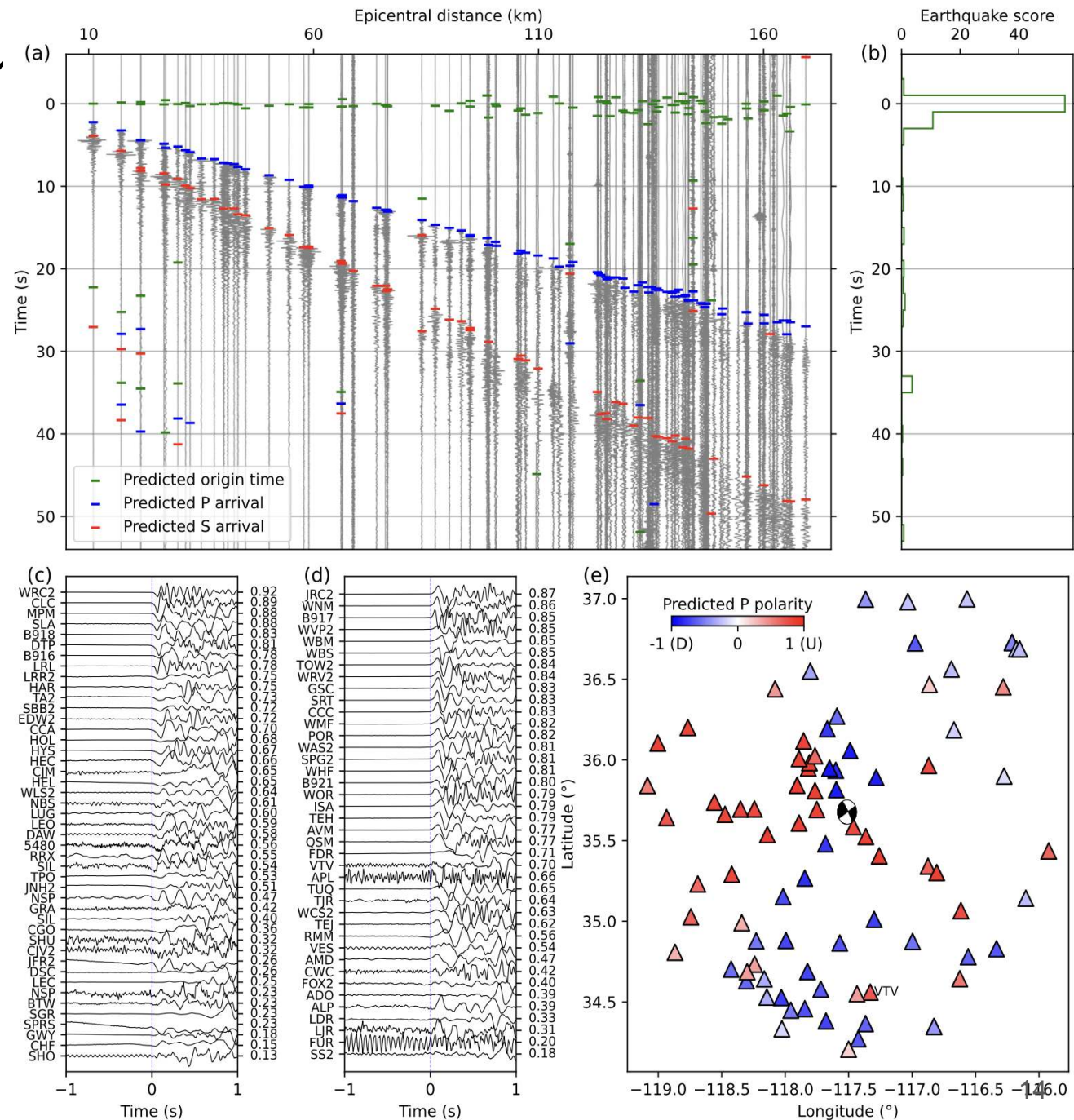
Improvement: simple association based on origin time

PhaseNet+: Multi-task Learning

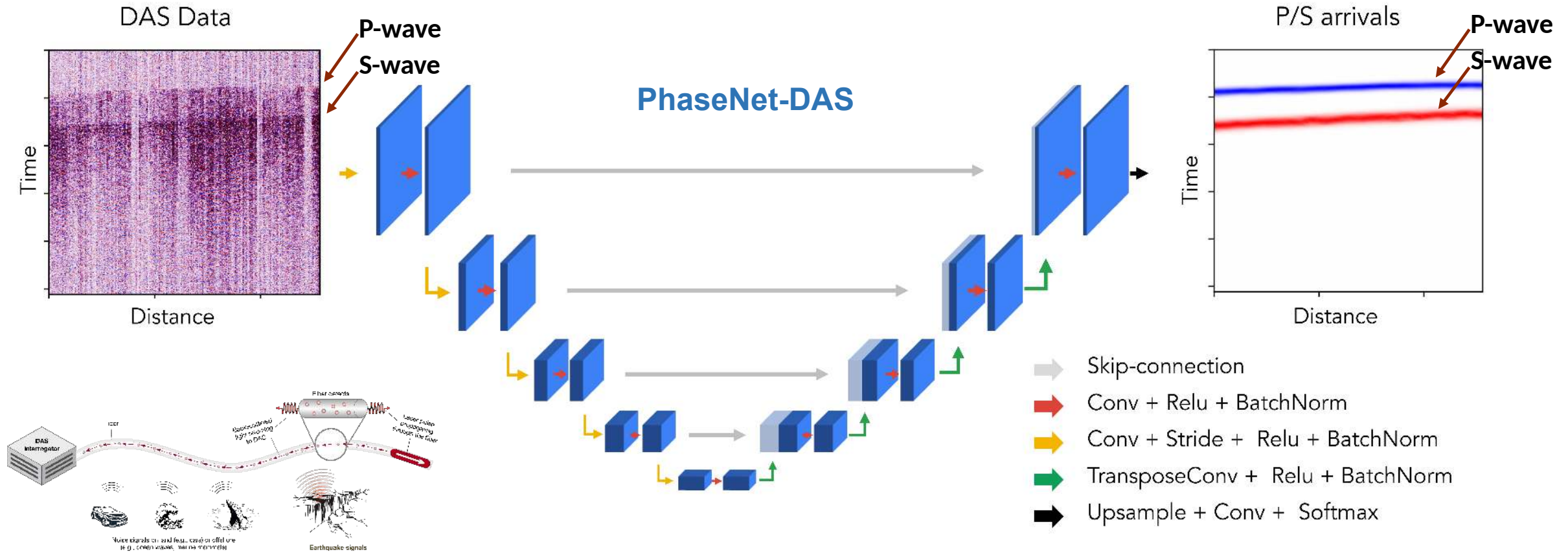


P&S Phase Arrival-time
+ Phase Polarity
+ Origin Time

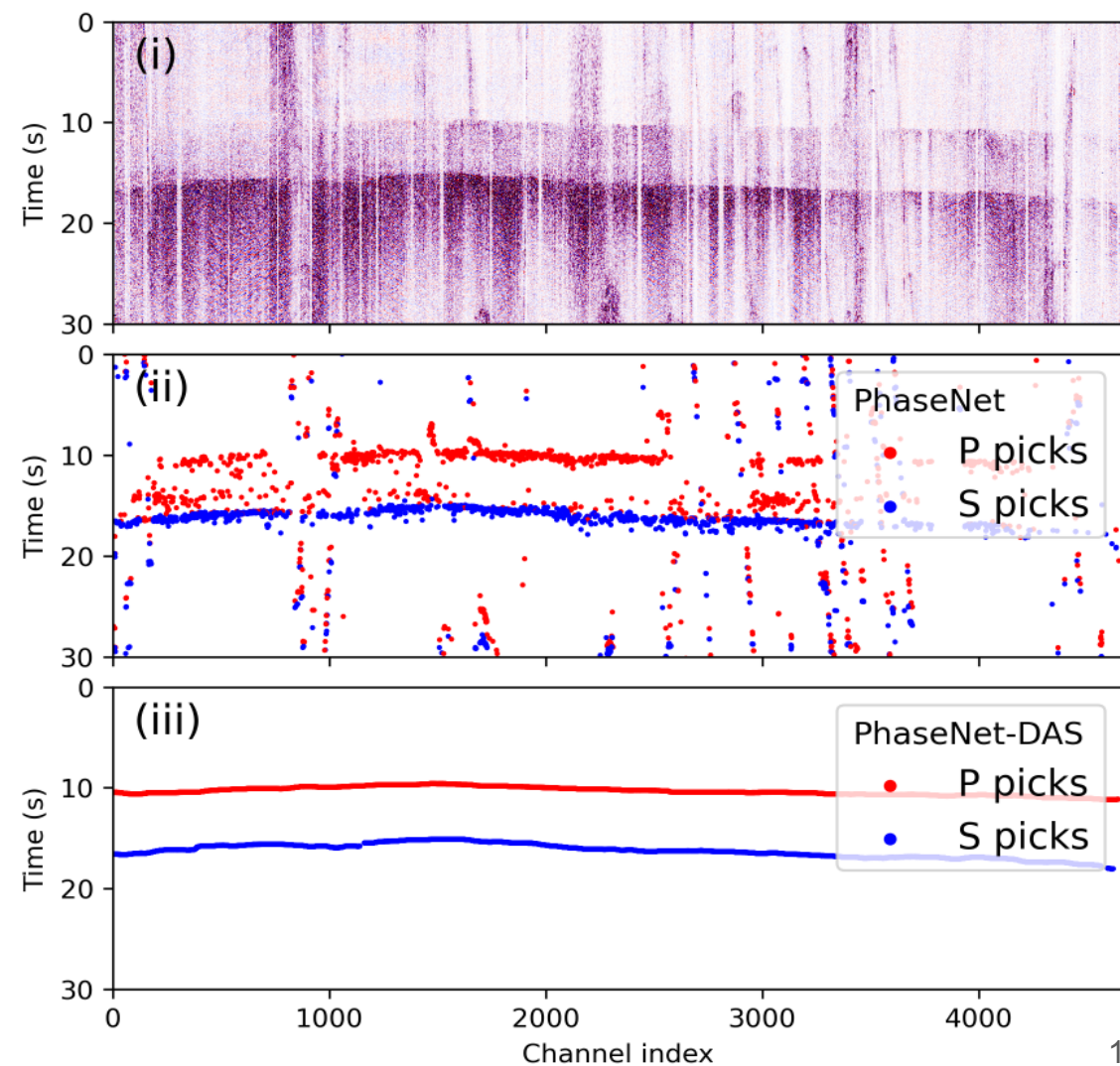
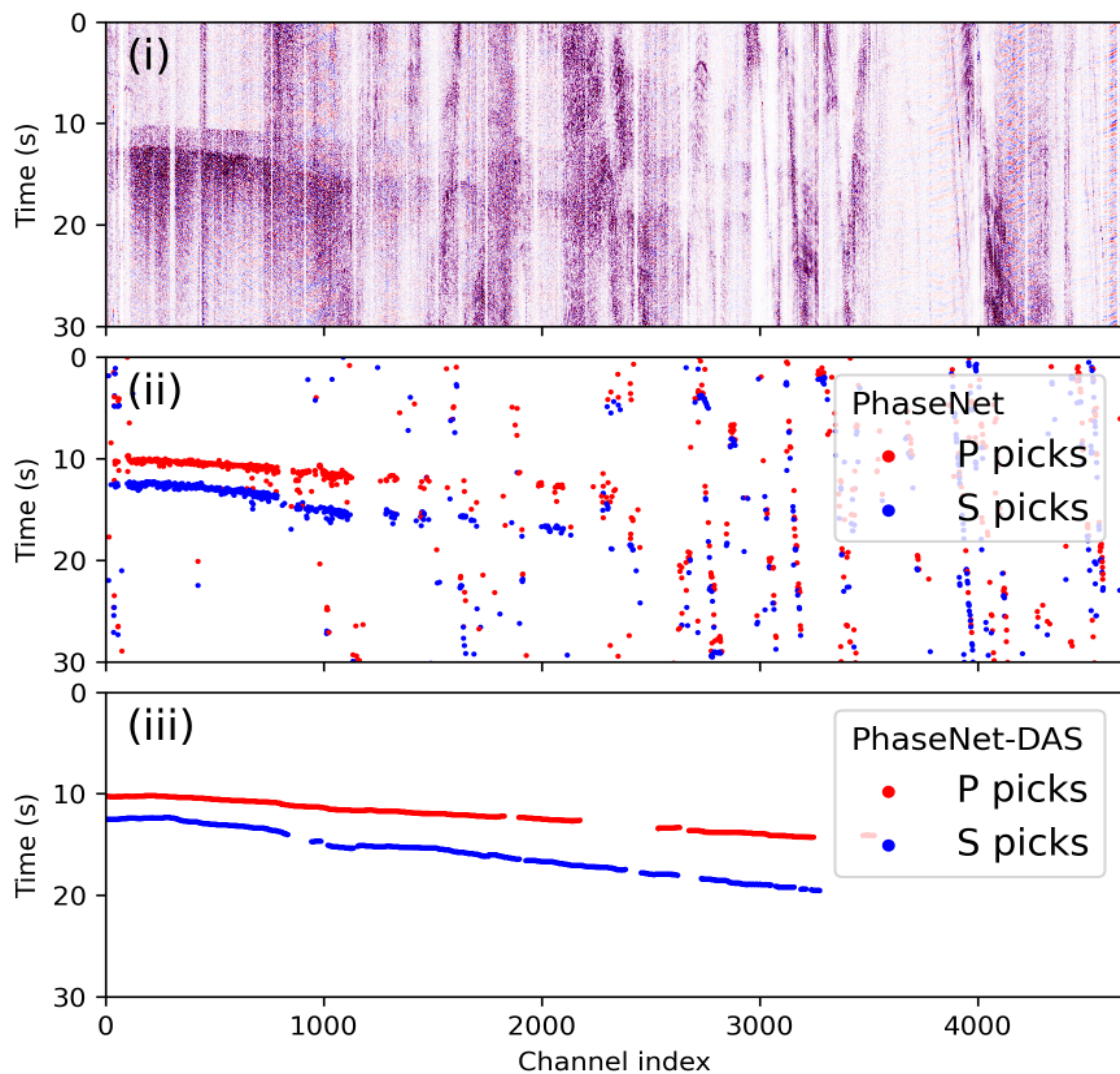
(Zhu et al.,
2025)



PhaseNet-DAS: Phase Picking for DAS



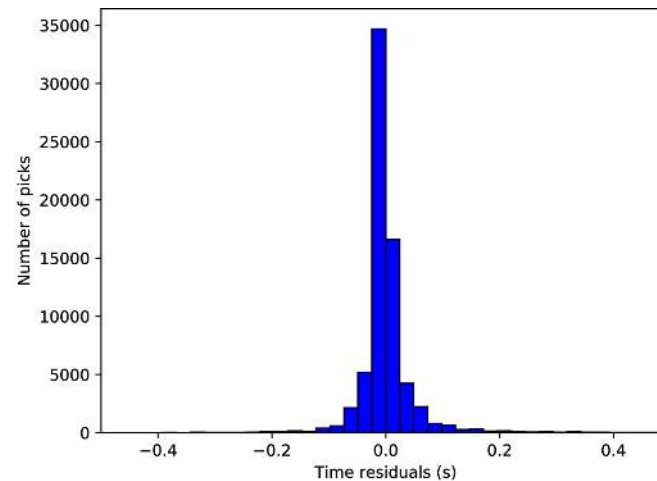
Examples:



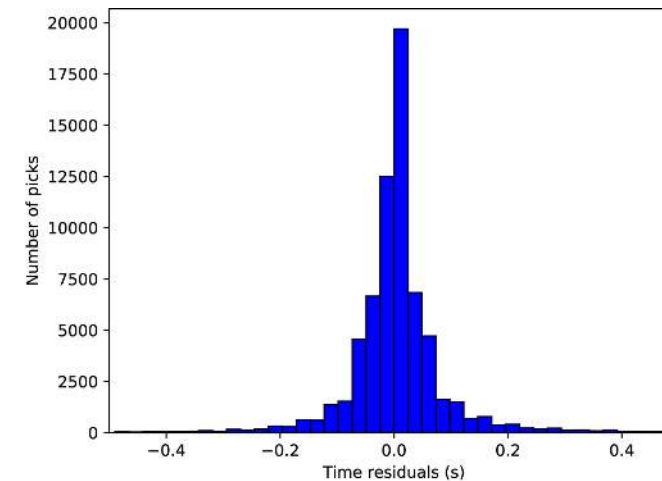
Phase Time Errors

PhaseNet

(compared with manual labels)



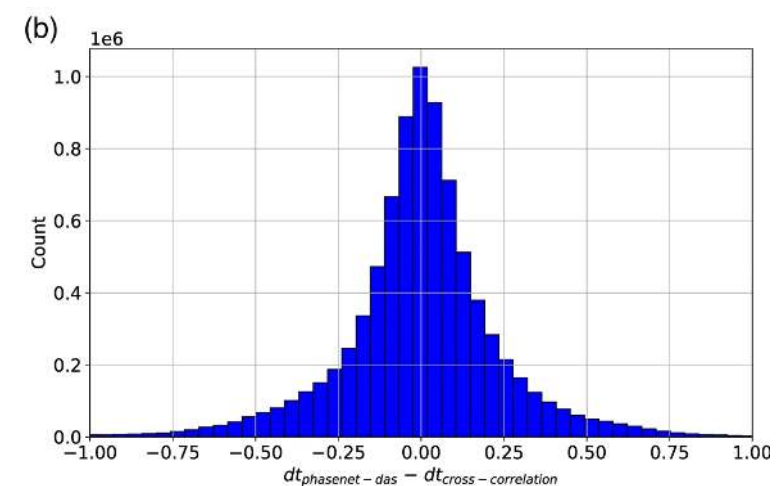
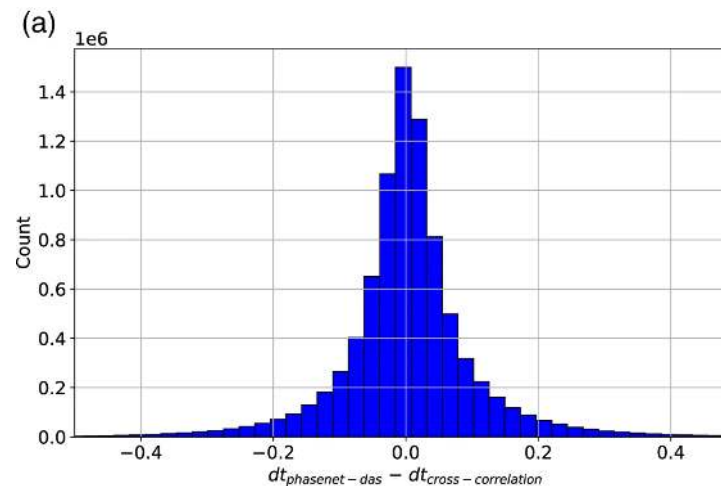
(a) P picks of PhaseNet



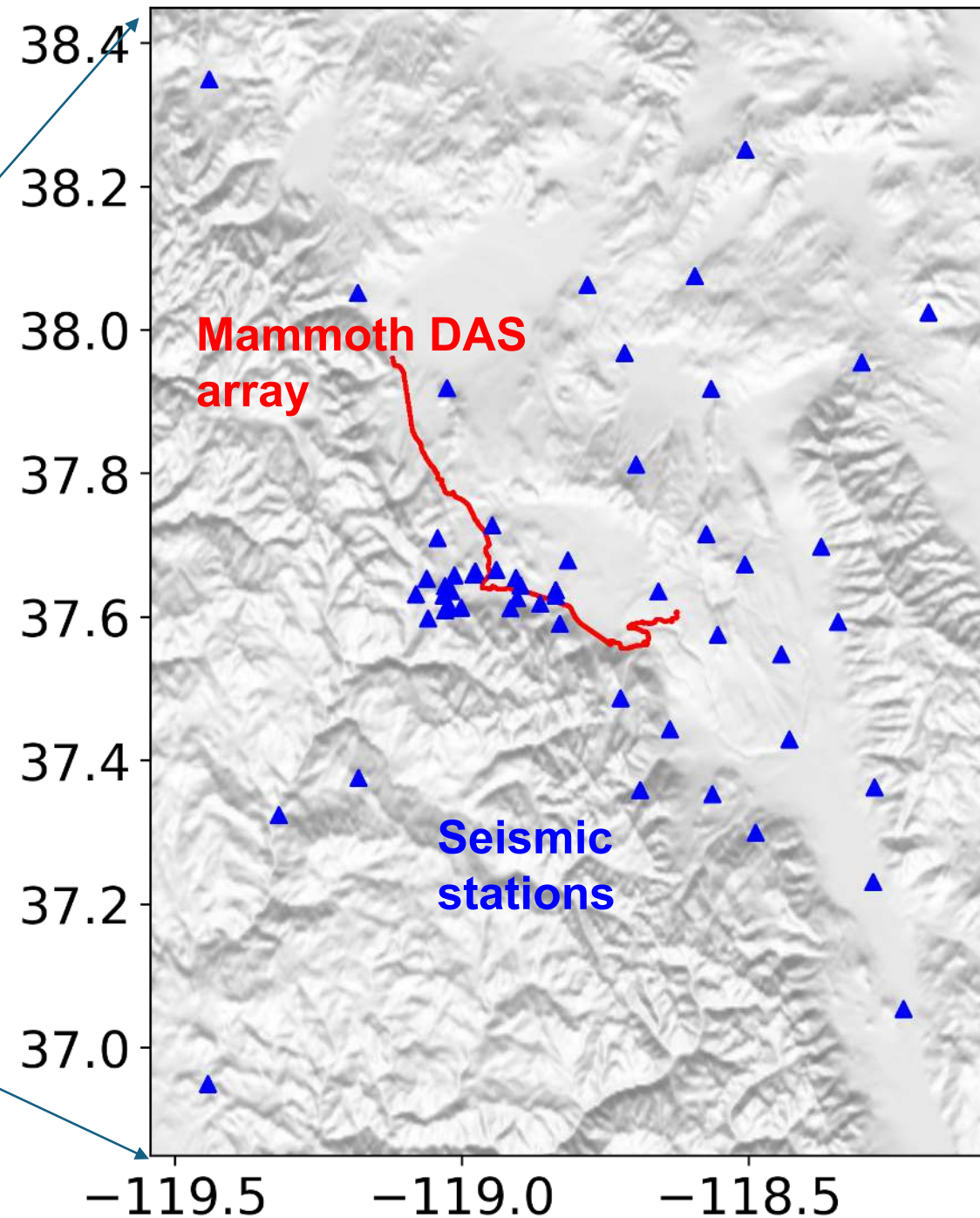
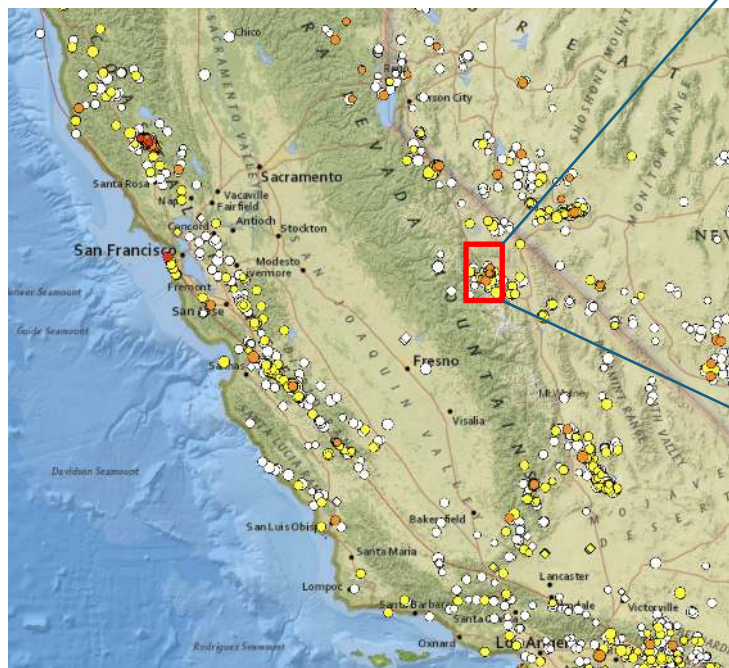
(b) S picks of PhaseNet

PhaseNet-DAS

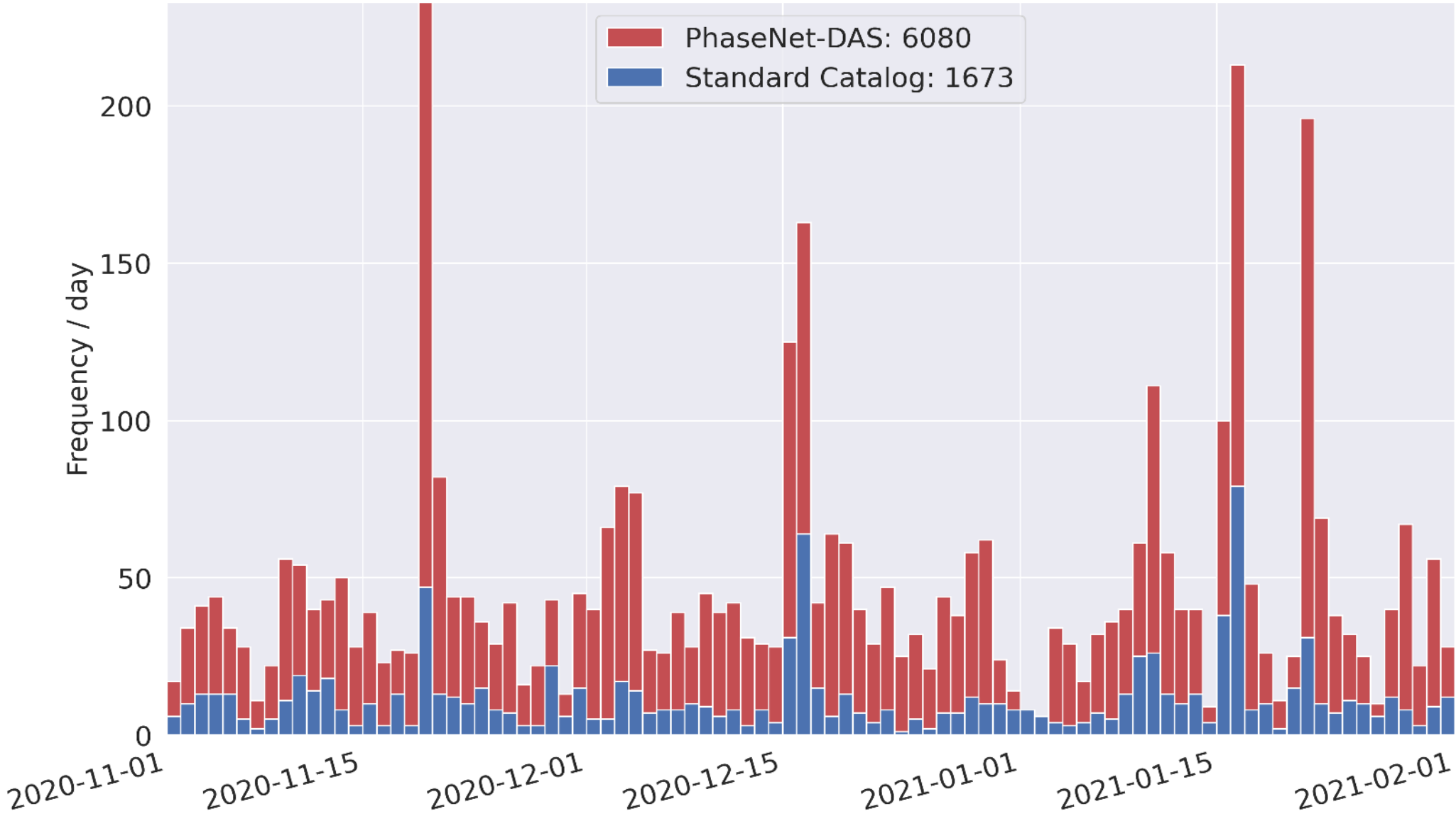
(compared with cross-correlations)



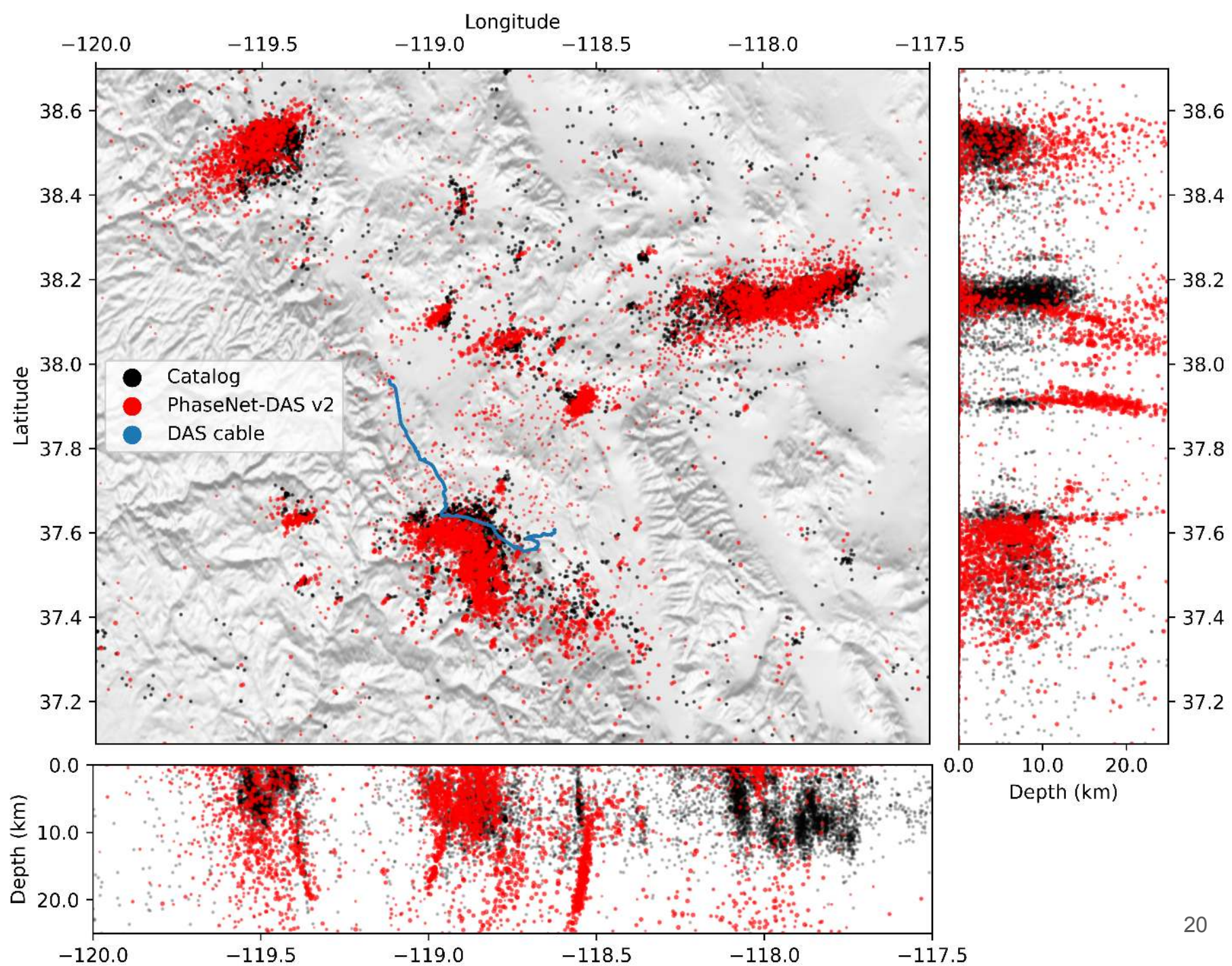
Seismic Network vs. DAS



Detected Earthquake Frequency

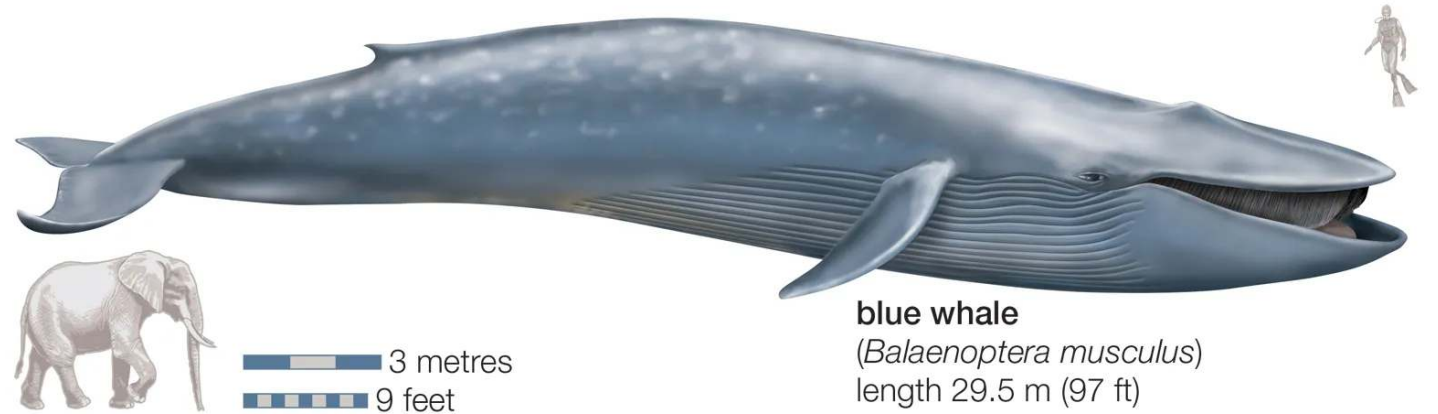


Approximate Earthquake Locations

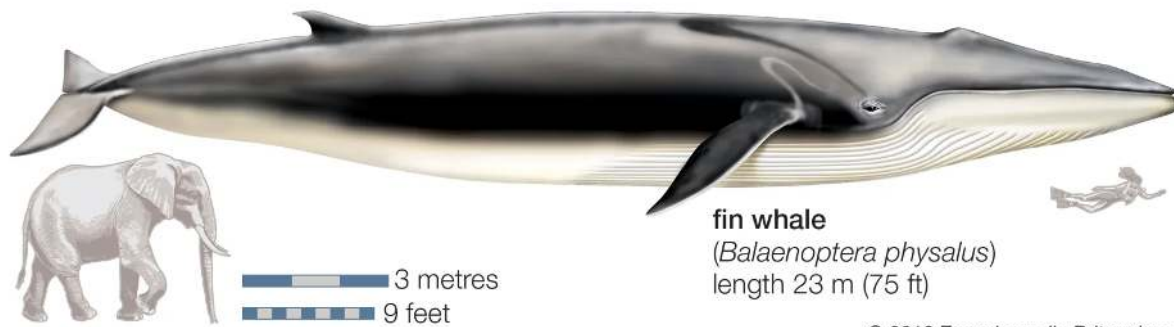


Beyond Earthquakes

How to automatically detect submarine DAS signals?

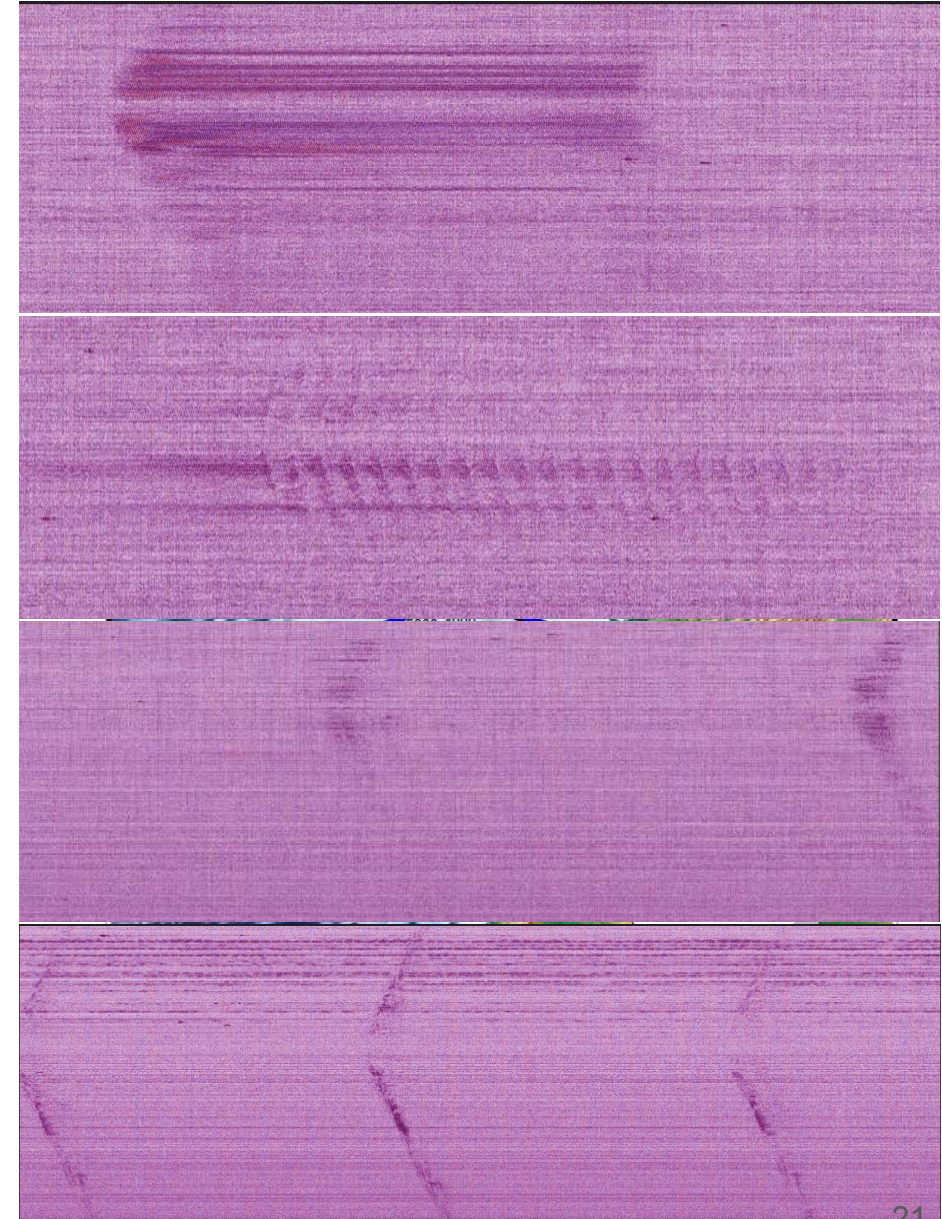


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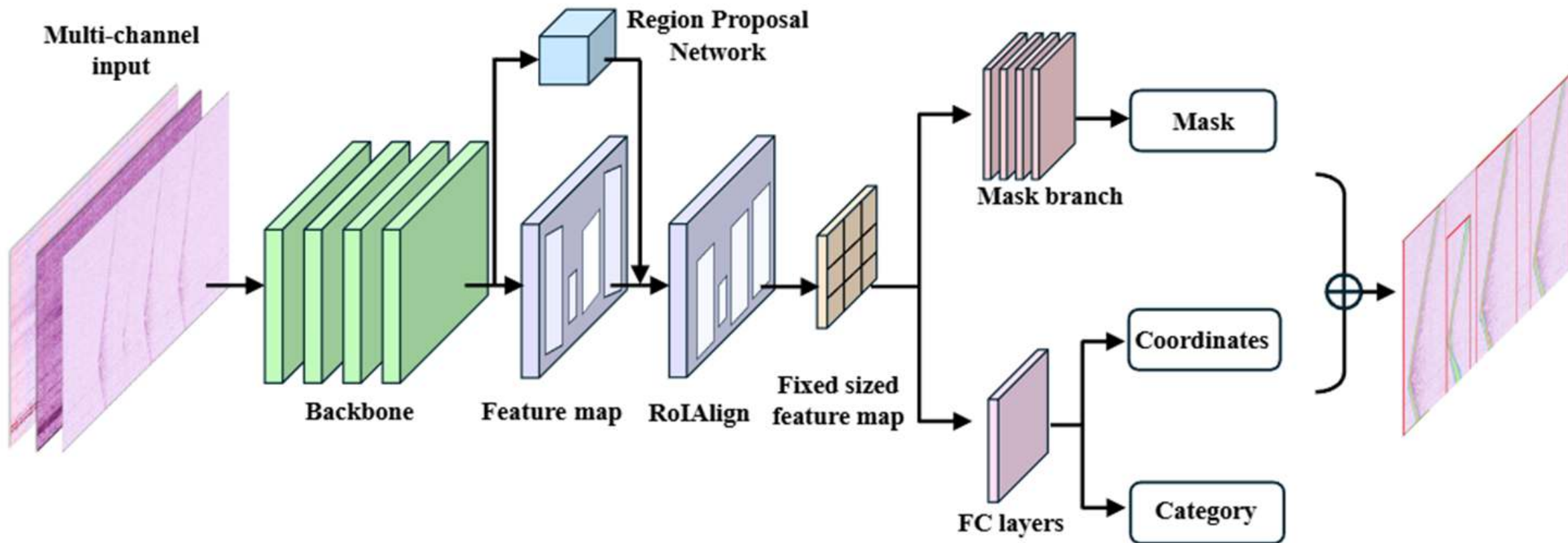


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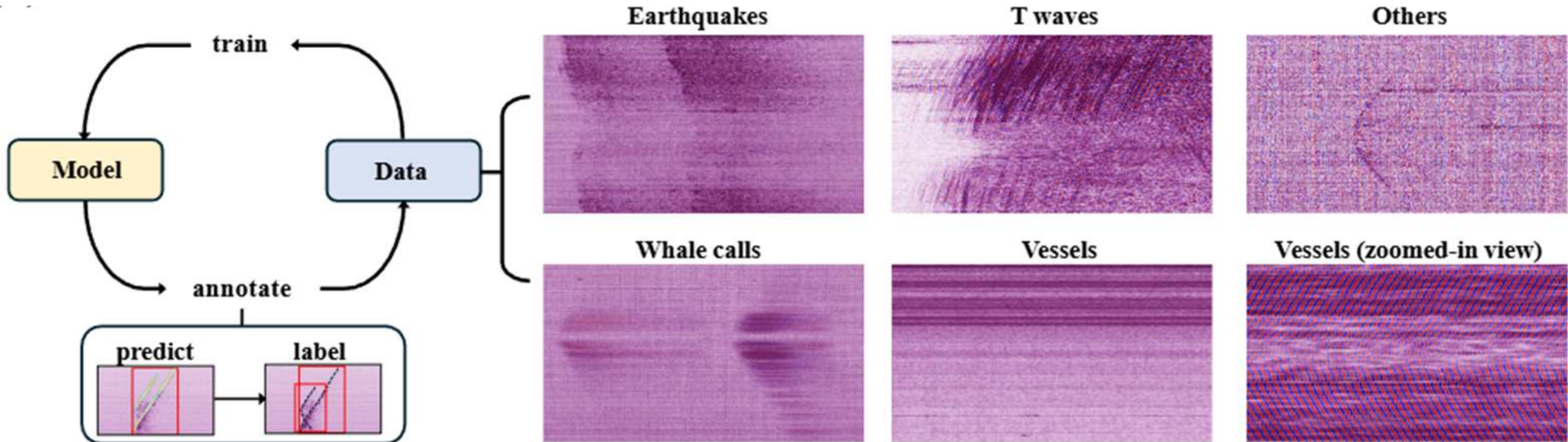
(Romanowicz et al., 2023)



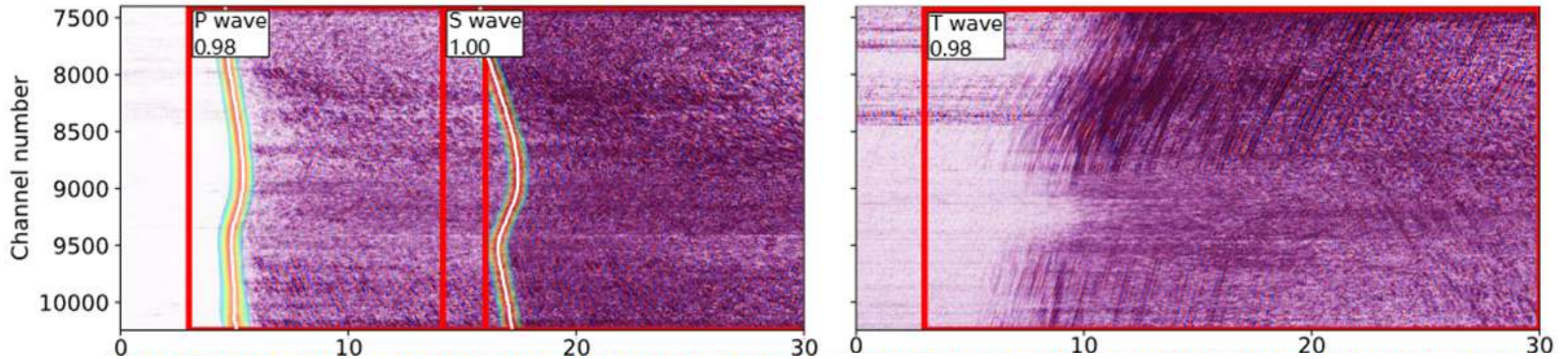
DASNet: Signal Detection & Picking



Semi-supervised Data Engine

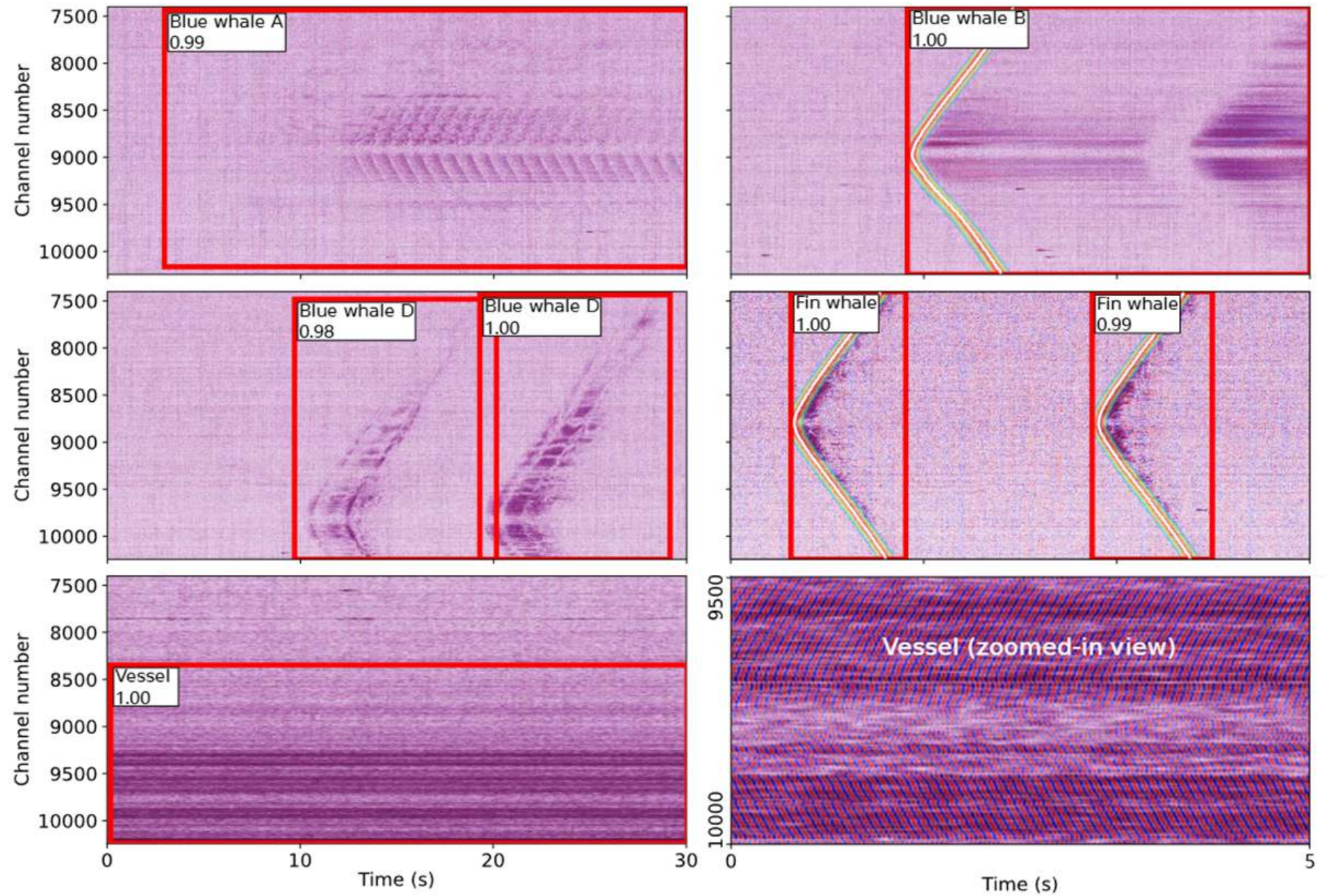


Examples:

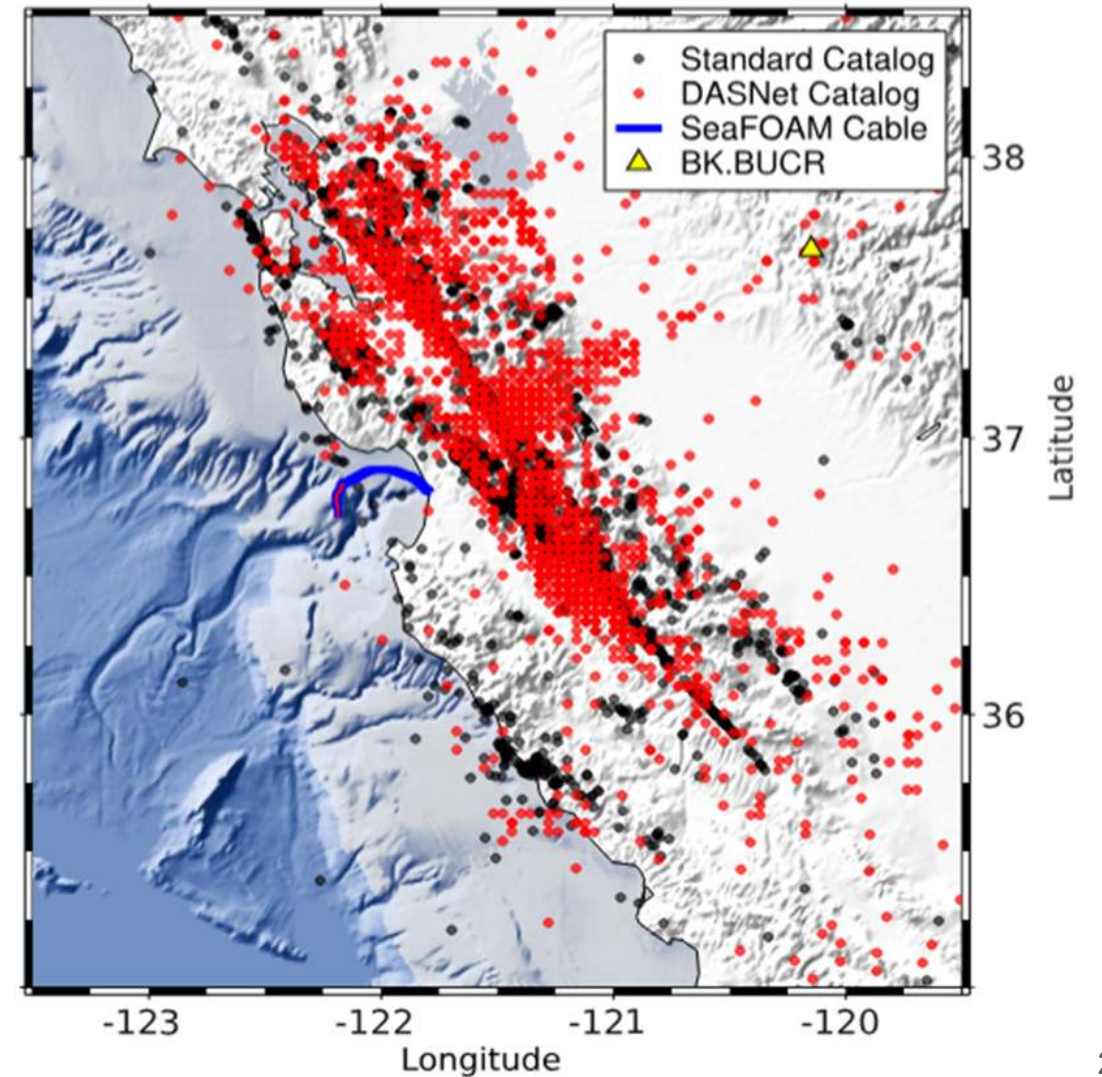
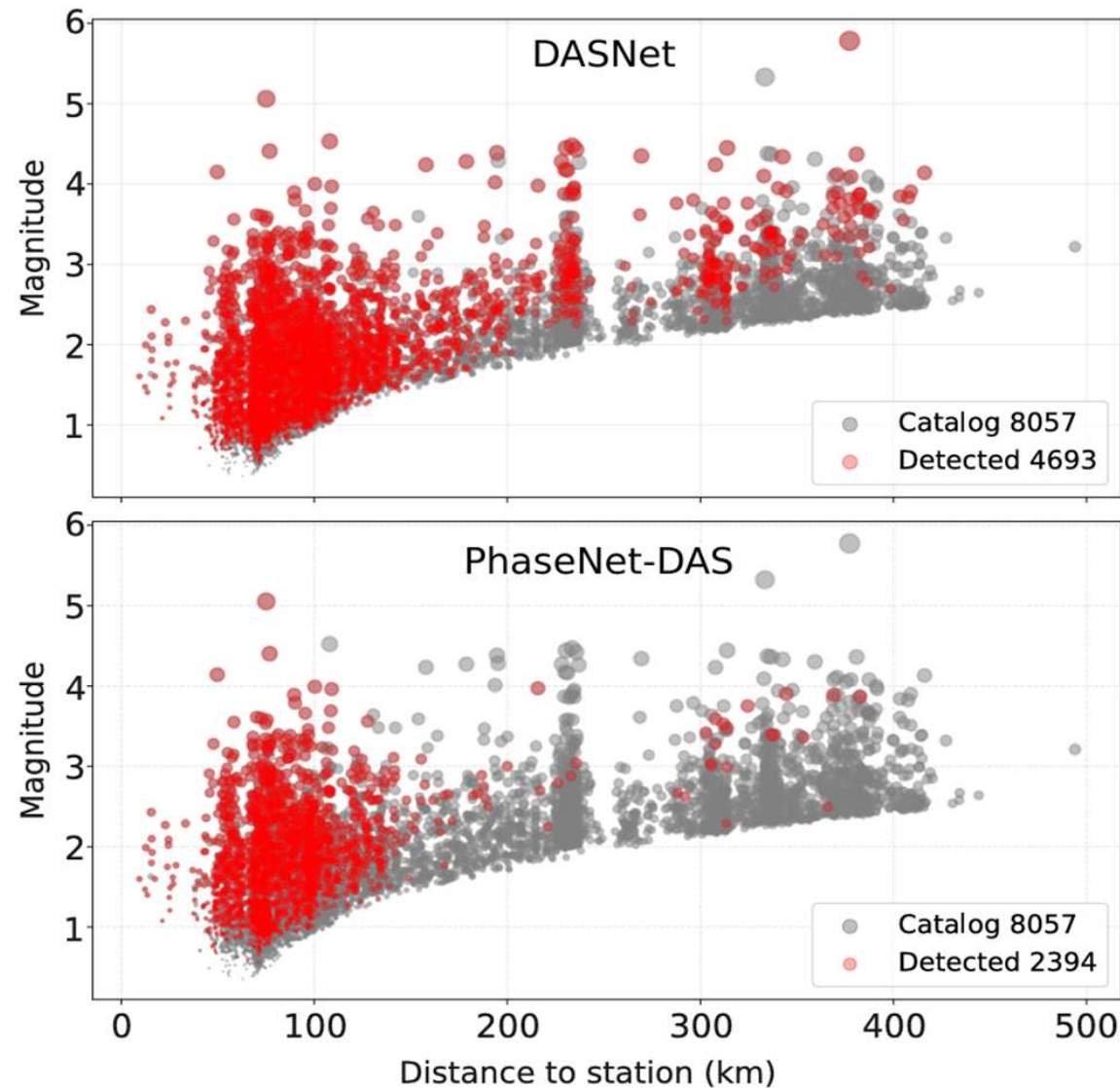


- Bounding box to represent the extent of the detected signal.
- Signal category with score to represent the model confidence.
- Probability distribution mask for arrival picks (only for signal with clear arrival).

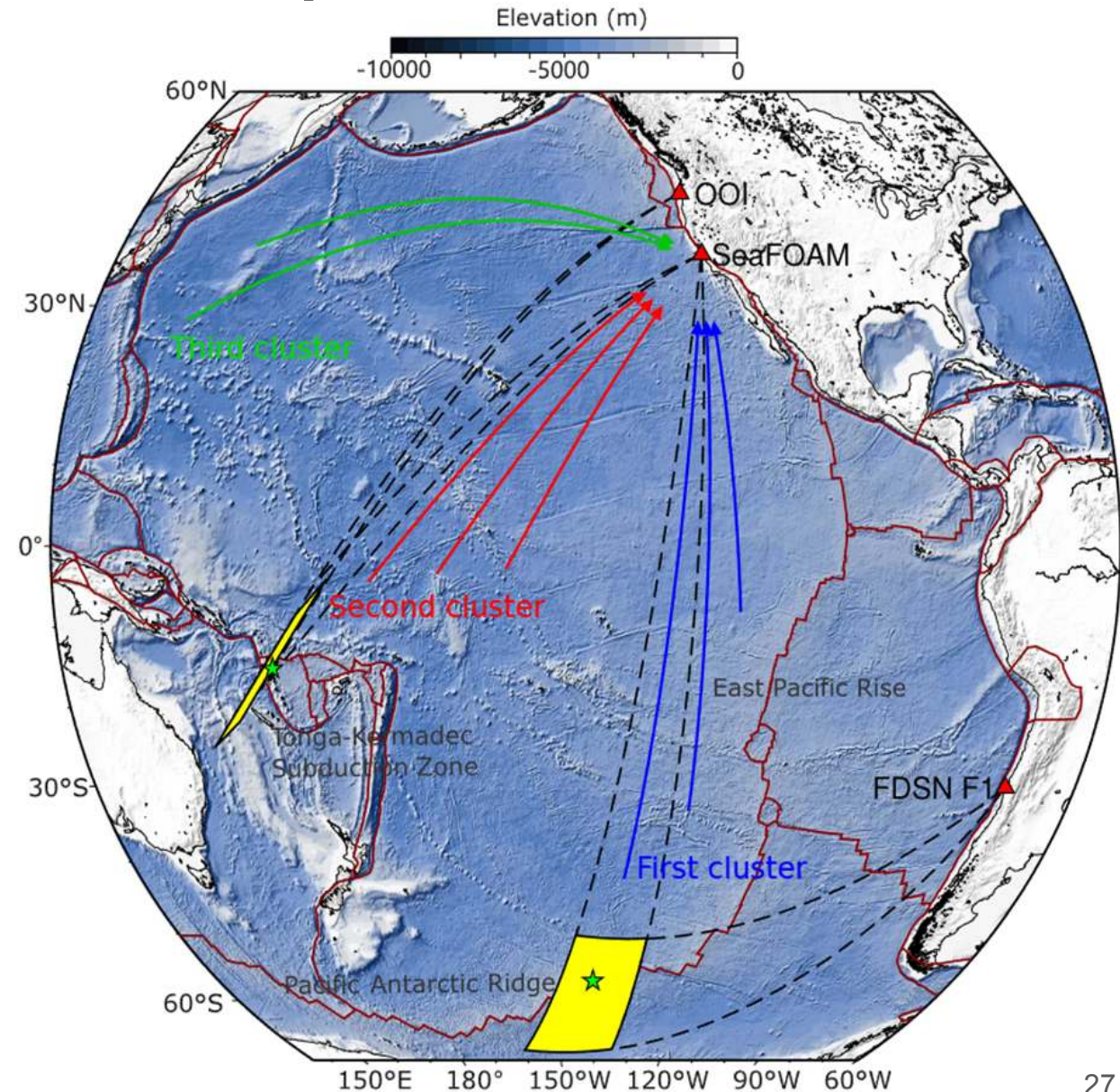
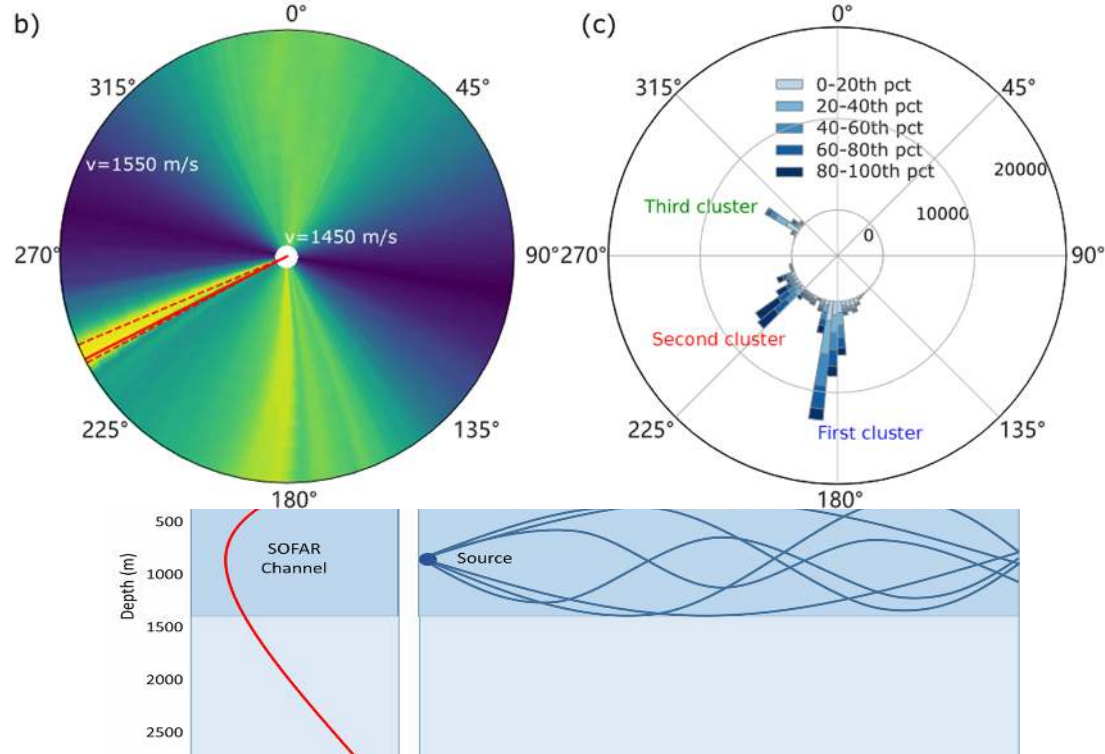
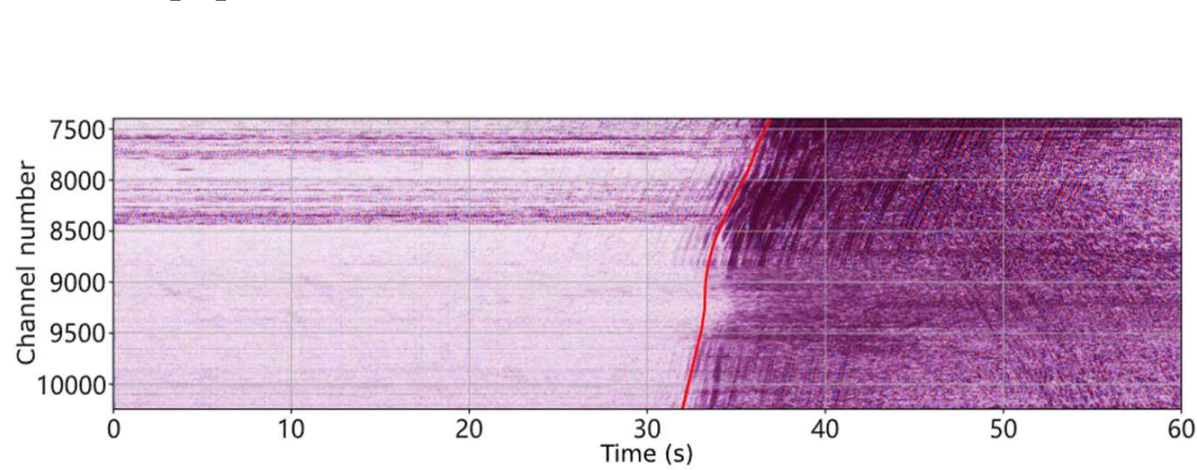
Examples



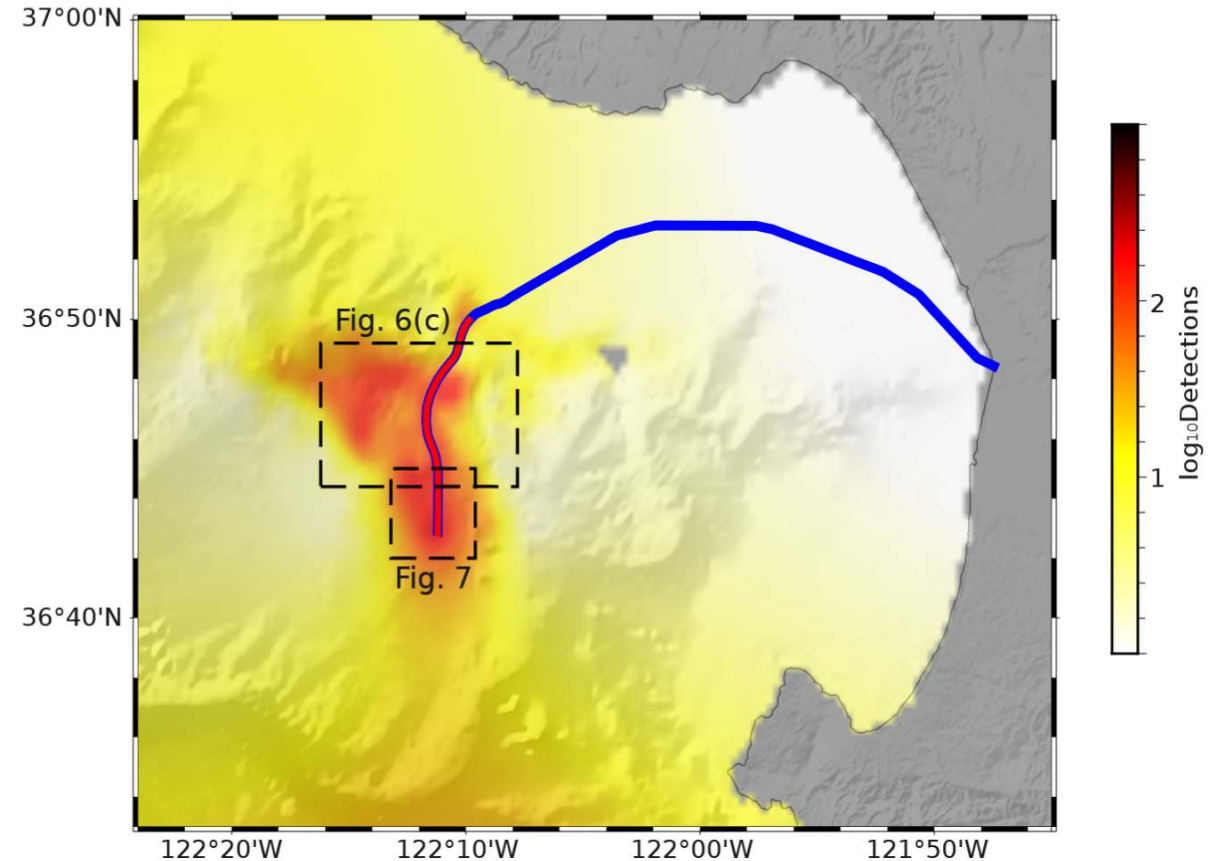
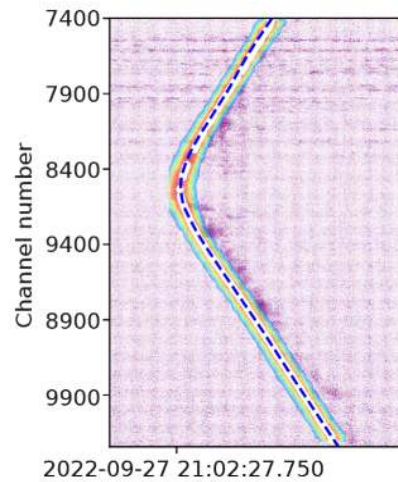
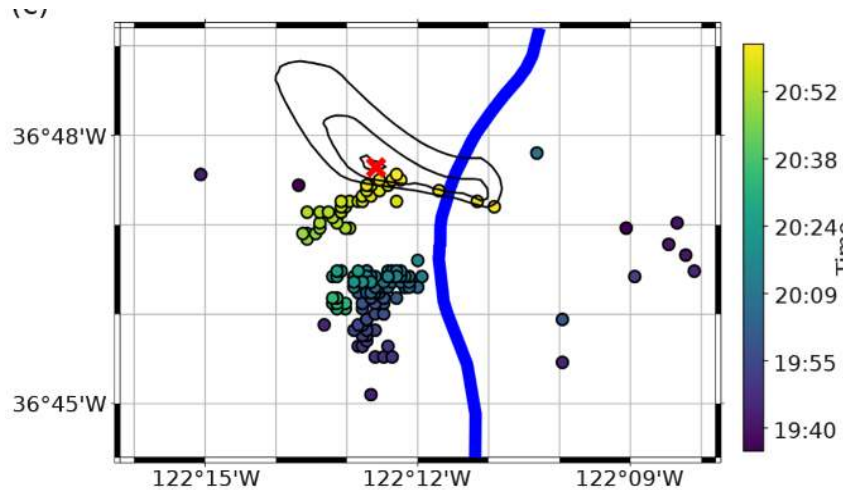
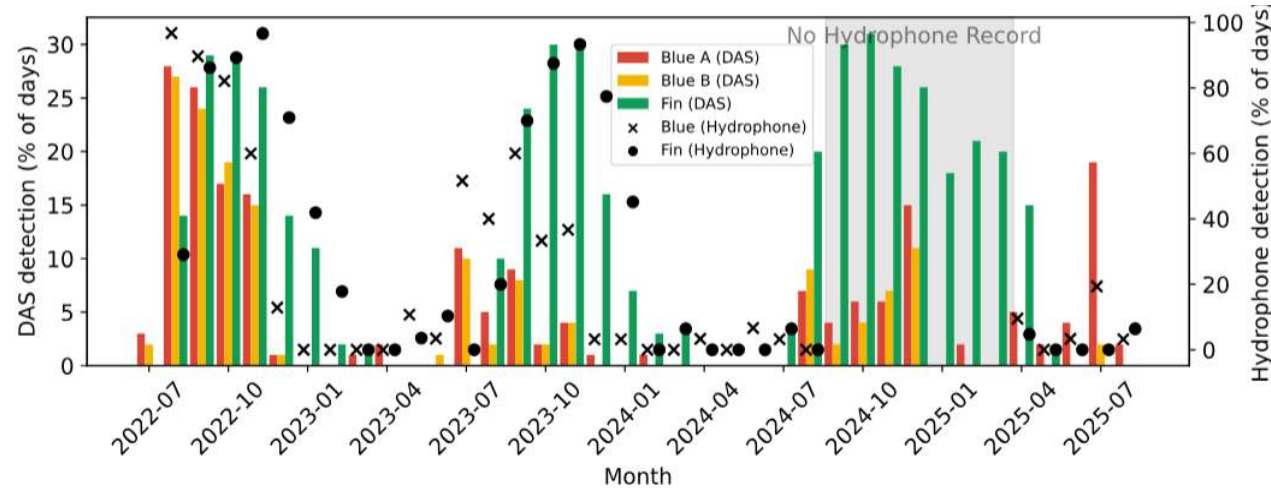
Applications: Earthquake Monitoring



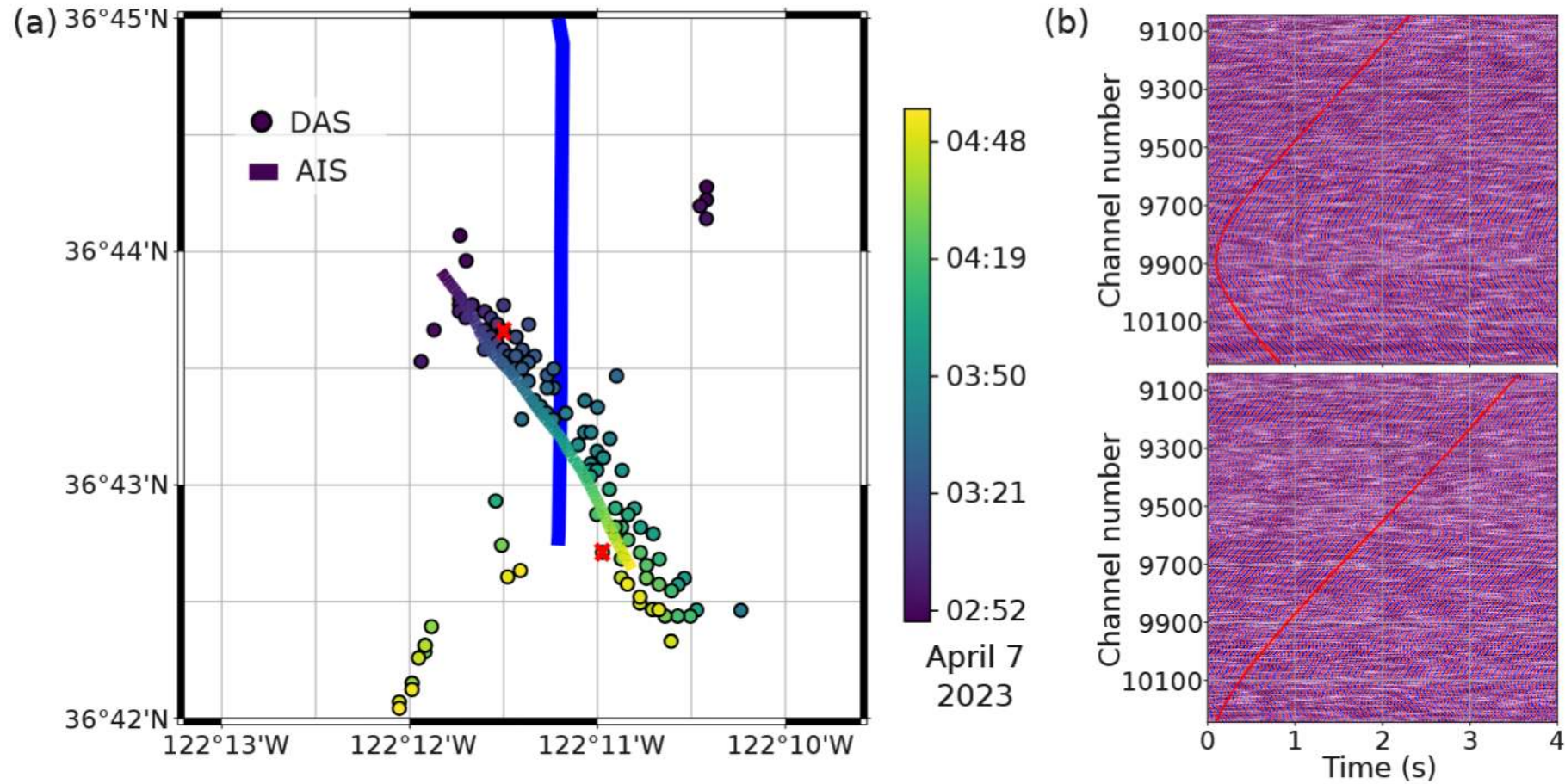
Applications: T-wave from Distant Earthquakes and Volcanoes



Applications: Tracking Bioacoustic (whale call) Sources

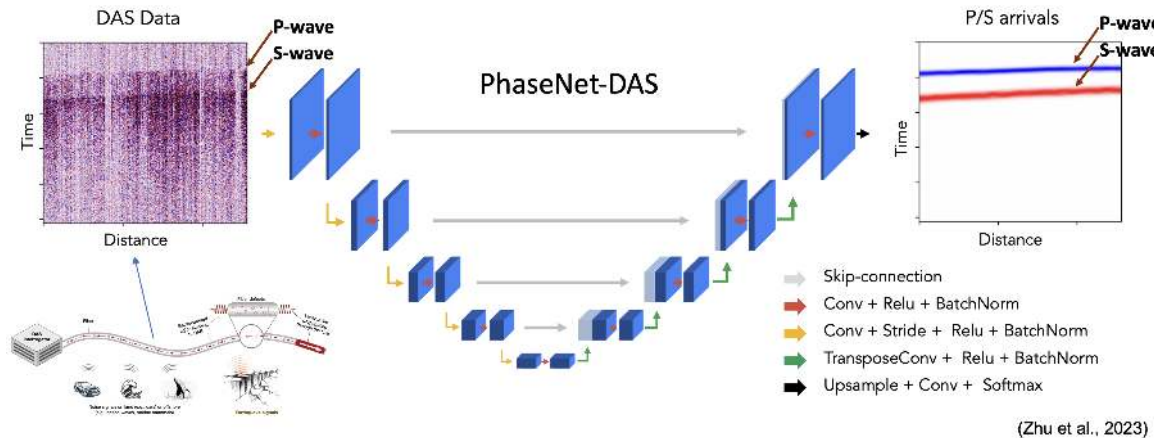


Applications: Tracking Anthropogenic (Vessel) Sources



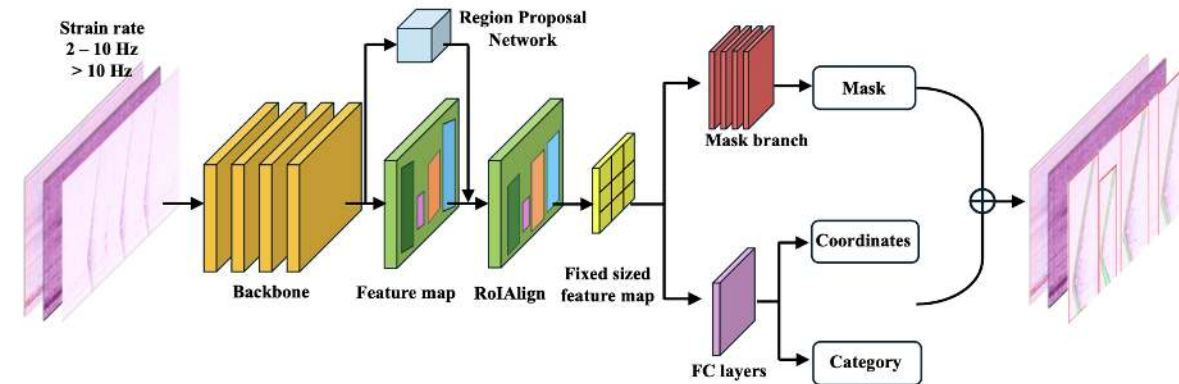
Deep Learning + DAS

- PhaseNet-DAS

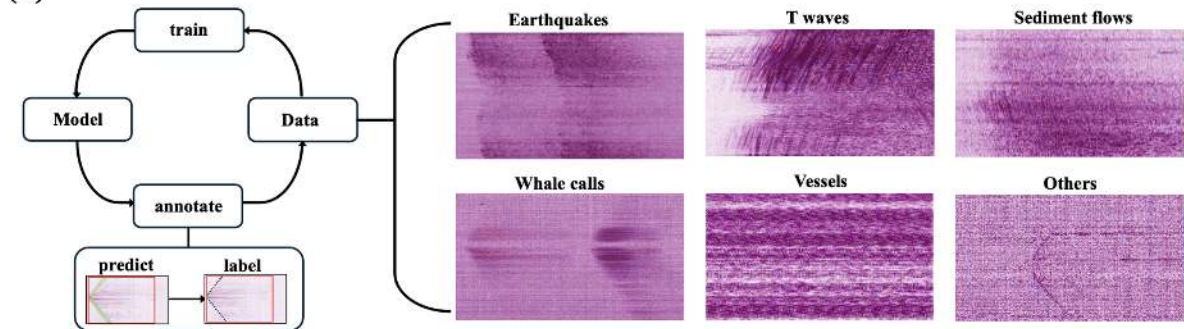


- DASNet

(a)



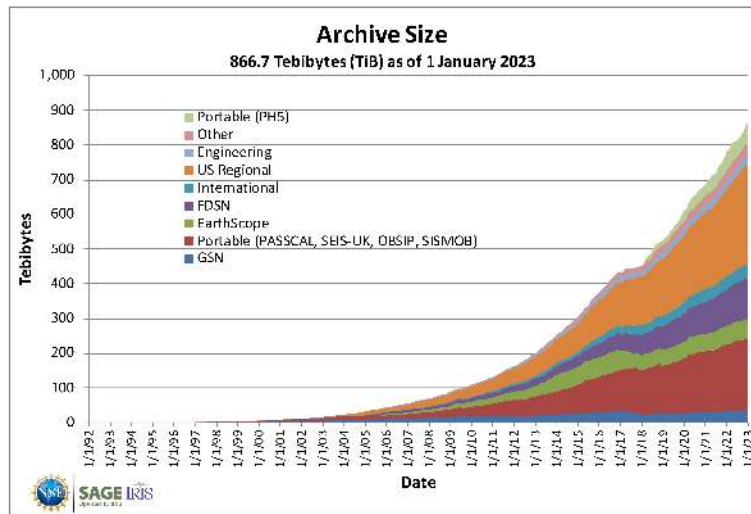
(b)



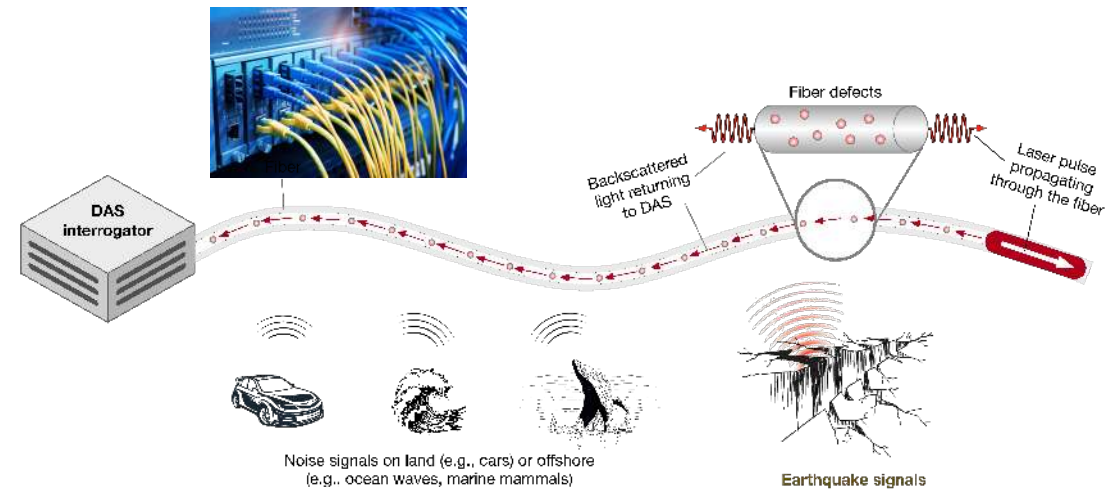
(Zhu et al., 2023; Zhang et al., 2026)

Big Data & Deep Learning in Seismology

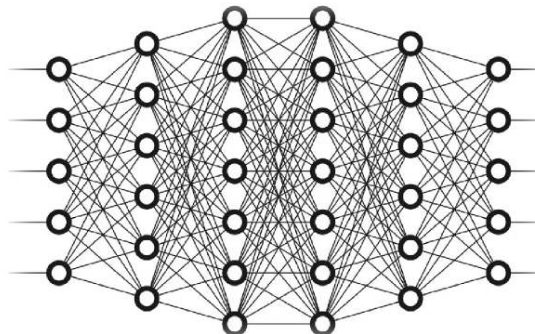
Seismic Data (Large T)



DAS Data (Large N)



Deep Neural Networks



Acknowledgements

We thank Google X, the California Broadband Cooperative, and the Los Angeles Department of Water and Power (LADWP) for providing access to telecommunication fibers.

We thank OptaSense and Silixa for their technical support.

We thank researchers for making the following datasets available:

- The GorDAS experiment in Arcata, CA (McGuire et al., 2025; USGS)
- The SeaFOAM experiment on the MARS cable in Monterey Bay (Romanowicz et al., 2023; BSL),
- The Ridgecrest DAS array via the SCEDC (<https://scedc.caltech.edu/data/cloud.html>).

We thank the Google Cloud Team for their support in running the workshop on the GCP Kubernetes cluster, and the Doerr School of Sustainability Computational Team for their help with the GCP setup.

We thank the SSA team for the opportunity to teach this workshop.

Related Papers:

- Mousavi, S. M., Sheng, Y., Zhu, W., & Beroza, G. C. (2019). STanford EArthquake Dataset (STEAD): A Global Data Set of Seismic Signals for AI. *IEEE Access*, 7, 179464-179476.
- Zhu, W. & Beroza, G. C. (2019). PhaseNet: A Deep-Neural-Network-Based Seismic Arrival-Time Picking Method. *Geophysical Journal International*, 216(1), 261-273.
- Zhu, W., McBrearty, I. W., Mousavi, S. M., Ellsworth, W. L., & Beroza, G. C. (2022). Earthquake Phase Association Using a Bayesian Gaussian Mixture Model. *Journal of Geophysical Research: Solid Earth*, 127(5), e2021JB023249.
- Romanowicz, B., Allen, R., Brekke, K., Chen, L., et al. (2023). SeaFOAM: A Year-Long DAS Deployment in Monterey Bay, California. *Seismological Research Letters*, 94(5), 2348-2359.
- Zhu, W., Biondi, E., Li, J., Yin, J., Ross, Z. E., & Zhan, Z. (2023). Seismic Arrival-Time Picking on Distributed Acoustic Sensing Data Using Semi-supervised Learning. *Nature Communications*, 14, 8192.
- Gou, Y., Allen, R. M., Zhu, W., Taira, T. A., & Chen, L. W. (2025). Leveraging Submarine DAS Arrays for Offshore Earthquake Early Warning. *Bulletin of the Seismological Society of America*, 115(2), 516-532.
- Zhu, W., Wang, H., Rong, B., Yu, E., et al. (2025). California Earthquake Dataset for Machine Learning and Cloud Computing. *arXiv:2502.11500*.
- Zhu, W., Song, J., Wang, H., & Münchmeyer, J. (2025). Towards End-to-End Earthquake Monitoring Using a Multitask Deep Learning Model. *arXiv:2506.06939*.
- Zhang, C., Zhu, W., Romanowicz, B. A., Allen, R. M., Soga, K., & Wu, Y. (2026). DASNet: A Deep Learning Framework for Marine Acoustic and Seismic Monitoring with DAS. *arXiv:2603.14844*.

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